Learning and compressing the product-formula residual: a Lie-algebraic spectral truncation that advances the Hamiltonian-simulation error–cost frontier

Molena Huynh[1]

[1] ***North Carolina State University*** · ***Correspondence:*** `molena.huynh@jmp.com`

Abstract

Digital Hamiltonian simulation by Trotter–Suzuki product formulas trades accuracy for circuit depth: reaching a smaller error demands a higher-order formula and many more gates. We show that this trade-off can be relaxed by treating the finite-step error itself as a structured, compressible, learnable operator. The residual factor $R_q = U(\delta t)S_q(\delta t)^\dagger$ of an order-q step is the unique unitary correction to the exact propagator and is optimal in every unitarily invariant norm; we prove that projecting its Hermitian generator onto the Pauli-weight filtration of the Lie algebra of Hermitian operators is the unique Frobenius-optimal compression—a *Lie-algebraic spectral truncation* whose level is a single knob interpolating from the uncorrected step to the exact correction. We prove this truncation converges *geometrically* in the level—a measured per-level error reduction of 4.5–5.8×10^2—and carries an a priori certificate $r\,\|K_q - \Pi_w K_q\|_2$ that fixes the truncation bandwidth directly from the target accuracy, before any simulation. Because the residual generator is geometrically local at every order (leading Pauli weight at most $q+2$) it compiles faithfully into mutually-commuting two-qubit-gate layers, and a symmetric inner compilation removes the low-order compilation floor at every order. The downstream payoff is a genuine advance of the product-formula error–cost frontier: a spectrally-truncated fourth-order correction reaches global spectral-norm errors lying between standard sixth- and eighth-order Suzuki formulas at 4.2–5.2× fewer two-qubit gates than the cheapest standard formula of equal accuracy ($n = 5$–7), placing new Pareto-optimal points on the frontier. The truncation is learnable without the dense propagator: a single translation-equivariant network maps local couplings to the truncated generator's coefficients and, trained only on four- and five-qubit chains, transfers to ten qubits ($R^2 = 0.9999$). The mechanism extends from the transverse-field Ising model to the XXZ chain. We state the scope precisely—the gain is a new accuracy regime reachable at a fraction of the next order's cost, not a uniform cost reduction—and every reported number is an exact, deterministic, reproducible dense-matrix computation.

Digital Hamiltonian simulation asks for accurate implementations of $U(t) = \exp(-iHt)$ for a many-body Hamiltonian H on a finite-dimensional Hilbert space. Since Lloyd's observation that local quantum systems can be simulated efficiently by quantum computers [1], product formulas have remained one of the most transparent and hardware-aligned simulation strategies. The basic Trotter product formula originates in Trotter's theorem [2]; higher-order recursive splittings were developed by Suzuki [3–5] and by related symplectic-composition methods [6–10]. Product formulas also underlie early quantum simulation proposals [11–13], phase-estimation workflows [14–17], and practical resource estimates in quantum chemistry and lattice simulation [18–24].

The current theory of product-formula simulation is much sharper than the original asymptotic picture. Commutator-sensitive bounds explain when product formulas outperform worst-case estimates [25–27], locality improves lattice-simulation scaling [28–30], and the cost of implementing Trotter steps has itself become an object of complexity analysis [31]. At the same time, product formulas compete with multi-product formulas [32], truncated Taylor methods [33], quantum signal processing and qubitization [34–37], randomized formulas [38–41], Magnus-expansion algorithms [42–45], variational fast-forwarding [46], and Lie-algebraic decomposition or compilation approaches [47–50]. These methods are complementary, but they all confront the same conceptual object: the finite-step defect between the desired unitary and the implemented unitary.

This work builds most directly on the Hamiltonian-simulation neighborhood of the second-order Zassenhaus factorization and the associated root-activity complexity bounds [71, 72]. We adopt the second-order Zassenhaus generator as an analytic prior on top of which a network learns a structured residual correction, and we prove how that correction's approximation error propagates across steps. The contribution is complementary to changing the product formula or proving a representation-theoretic lower bound: by compressing and faithfully compiling the residual generator of a *higher-order* step, we move the product-formula accuracy–cost frontier itself (Sec. 1.7), under guarantees inherited from the analytic prior and the locality of the defect.

We isolate that defect as an operator to be studied, bounded, compressed, and ultimately compiled. For a product-formula step $S_q(\delta t)$ of order q, define

$$R_q(\delta t) = U(\delta t) S_q(\delta t)^\dagger, \tag{1}$$

where $U(\delta t) = \exp(-iH\delta t)$. The corrected step $R_q(\delta t)S_q(\delta t)$ is exactly $U(\delta t)$ in exact arithmetic. On its face this identity is simple; the point is to turn the identity into a rigorous benchmark and a nontrivial approximation problem. The residual is unique, unitary, norm-optimal, and stable under approximation. Its Hermitian generator, defined by $R_q(\delta t) = \exp[-iK_q(\delta t)]$, is small for small δt and inherits local commutator structure from the product-formula defect. Once K_q is exposed, a natural compressed implementation problem emerges: choose an implementable operator subspace $\mathcal{B}$, project or learn K_q inside that subspace, and implement $\exp[-i\Pi_{\mathcal{B}} K_q] S_q$ instead of S_q.

We distinguish two modes of residual-generator Trotter compilation (RGTC). *Oracle RGTC* computes R_q from the exact dense matrix exponential. This mode is not scalable and is not claimed to be a deployable quantum algorithm; it establishes the target and the best possible one-step left correction. *Compressed RGTC* uses a Hermitian generator $\widehat{K}_q$ from a restricted family to form $\widehat{R}_q = \exp(-i\widehat{K}_q)$. This mode is the practical object: it can be instantiated by Baker–Campbell–Hausdorff (BCH) truncation, Pauli-weight projection, variational circuit compilation, operator learning, or GPU-accelerated offline fitting. This separation is essential: the exact identity $U S_q^\dagger S_q = U$ is trivial on its own, and the substance lies in quantifying what happens when the residual is approximated and in showing that structured approximations work. We do both, and more: we compile the compressed mode into gates and show that, applied to an already-high-order step, it advances the product-formula error–cost frontier (Sec. 1.7), and we instantiate it as a *learned* residual generator that, on disordered Ising chains, transfers from small systems to larger ones it never saw, under the proven stability guarantee. The compressed-and-compiled mode is the central practical advance, the learned residual generator makes such corrections obtainable without the dense propagator, and the exact theory makes both principled.

Our contributions are as follows. (i) We formalize RGTC and prove exact residual factorization, uniqueness, and global norm optimality among all left corrections for every unitarily invariant norm. (ii) We prove multi-step stability theorems for approximate residuals and generator approximations, giving explicit $r\eta$ and $r\left\|K_q - \widehat{K}_q\right\|_2$ error targets, and prove the linear-in-r estimate is order-optimal. (iii) We recast Pauli-weight projection of the residual generator as a *Lie-algebraic spectral truncation*—the unique Frobenius-optimal compressed generator (Theorem 7)—whose level w interpolates from no correction ($w = 0$) to the exact generator ($w = n$), the Hamiltonian-simulation counterpart of the operator-algebraic spectral truncations recently introduced in kernel learning [73, 74]. (iv) We prove the truncation converges *geometrically* in the level—each excluded weight level carries residual content of strictly higher order in δt (Theorem 10)—with a measured per-level reduction of 4.5–5.8 $\times$ 10^2 (Fig. 3a, Table 3). (v) We prove the a priori certificate $r \, \|K_q - \Pi_w K_q\|_2$ is a valid global error bound at every level (Fig. 3c), so it selects the minimal truncation bandwidth from the target accuracy *before* any simulation (Corollary 2). (vi) We prove residual locality holds at every order: the leading order-q residual generator of the open-boundary transverse-field Ising model (TFIM) has Pauli weight at most $q+2$, with an effective threshold $\approx \lceil q/2 \rceil + 2$ (Table 2), so higher-order corrections stay compressible. (vii) We prove and verify that the truncated correction *compiles faithfully* into gates: because $K_q = O(\delta t^{q+1})$, a first-order product over mutually-commuting Pauli-rotation layers adds only $O(\delta t^{2(q+1)})$ error, negligible for $q \geq 4$ but *not* for $q = 2$, and a *symmetric inner compilation* removes this structural floor at every order ($O(\delta t^{3(q+1)})$ error, Theorem 12)—e.g. lowering the compiled $n = 6$, $q = 2$, $w \leq 4$ error from 2.18×10^{-5} to 3.76×10^{-7} (Table 4). (viii) This yields the central result: a genuine advance of the product-formula error–cost frontier. Under a standard two-qubit-gate (CNOT) cost model, spectrally-truncated fourth-order corrections reach global spectral-norm errors between standard sixth- and eighth-order Suzuki formulas at 4.2–5.2$\times$ fewer two-qubit gates than the cheapest standard formula of equal accuracy ($n = 5$–7), placing new Pareto-optimal points on the frontier (Fig. 4). (ix) We instantiate the learned mode: a single translation-equivariant network predicts the step-size-conditioned weight-≤ 3 coefficients of K_2 from local couplings of a disordered TFIM, transfers from four- and five-qubit training chains to ten qubits ($R^2 = 0.9999$ on held-out sizes), cuts the uncorrected Strang error by 58–86$\times$, is trained *without any dense* 2^n *oracle* from fixed-size local patches, and beats the analytic leading-order correction where the latter fails—all under the proven $r\eta$ stability bound. (x) We show the mechanism extends from the TFIM to the XXZ/Heisenberg chain, with a larger weight threshold (Table 7). (xi) We provide deterministic dense-matrix experiments, figures, tables, and code that recompute every reported number from first principles. Figure 1 summarizes the pipeline, from the local Hamiltonian to the certified, compiled, frontier-advancing correction.

Beyond quantum simulation, the central finding is of methodological interest to machine learning for the physical sciences: the truncation error of a numerical integrator, usually treated as an analytic object to be bounded, is here a *learnable* operator—geometrically local, conditioned on the discretization step, and transferable across system size—with its approximation quality controlled *a priori* by a proven stability bound rather than validated only *a posteriori*. The enabling compression is algebraic: the correction lives in the Lie algebra of Hermitian operators, and Pauli-weight projection is a *spectral truncation* of that algebra to a finite-dimensional, symmetry-respecting hypothesis class indexed by an interaction-range knob, in direct analogy with the operator-algebraic spectral truncations that interpolate a locality knob in C*-algebraic kernel learning [73, 74]. The analogy is exact at the level of the construction—a single parameter contracting a noncommutative algebra to a tractable, structure-respecting subspace—but the simulation setting supplies three guarantees the kernel construction does not: a quantitative *geometric convergence rate* in the level rather than mere convergence (Theorem 10); an *a priori dynamical certificate* that turns the truncation level into a function of the target accuracy (Corollary 2), whose own gate-level realization we repair by a symmetric compilation (Theorem 12), the analogue of the kernel construction's positivity-restoring smoothing; and a measurable downstream payoff—an advance of the algorithm's accuracy–cost frontier. This places the work within the broader programme of structure-preserving, physics-constrained operator learning, where a restricted, symmetry-respecting hypothesis class is what makes generalization both data-efficient and certifiable.

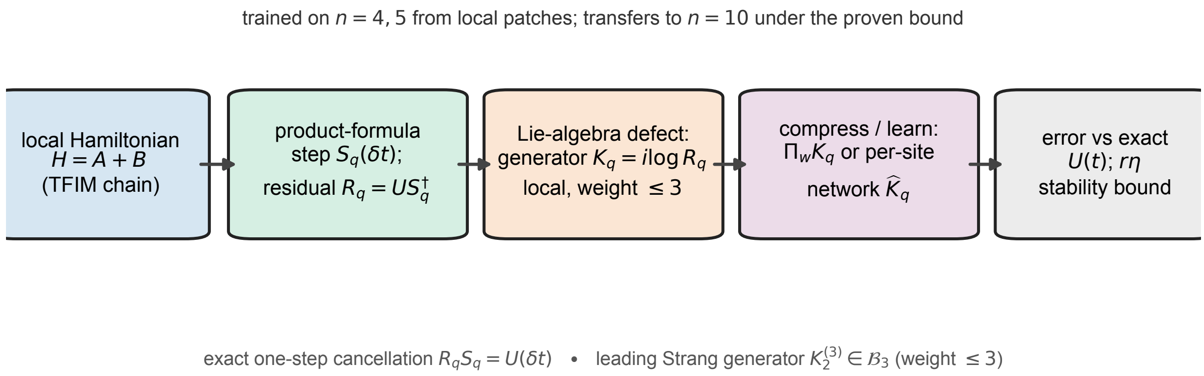

Figure 1. Overview of residual-generator Trotter compilation (RGTC). A local Hamiltonian $H = A+B$ is approximated by a product-formula step $S_q(\delta t)$; the exact residual factor $R_q = U(\delta t)S_q(\delta t)^\dagger$ defines a Hermitian residual generator $K_q = i \log R_q$ whose Lie-algebraic commutator structure makes the leading Strang term geometrically local (Pauli weight ≤ 3 for the TFIM; Theorem 8). This locality lets K_q be compressed by Pauli weight ($\Pi_w K_q$) or learned per site by a translation-equivariant network $\widehat{K}_q$ from local couplings, with no access to the dense propagator. The corrected step $\exp(-i\widehat{K}_q)S_q$ is scored by spectral-norm error against the exact $U(t)$, and inherits the proven multi-step stability certificate $\left\|\widehat{G}^r - U(t)\right\|_2 \leq r\eta$ (Theorem 4). The diagram is schematic and contains no numerical data.

1 Results

1.1 Hamiltonian model and product-formula baselines

We first fix the Hamiltonian model, the product-formula steps, and the error metric used throughout. Let $\mathcal{H} \cong \mathbb{C}^d$ and let $H = H^\dagger \in \mathbb{C}^{d\times d}$, with exact time-$t$ propagator $U(t) = e^{-iHt}$. We assume a decomposition $H = A + B$ with $A = A^\dagger$ and $B = B^\dagger$ each easier to exponentiate than their sum; this two-term setting covers the standard even-odd or diagonal-transverse splittings used in many spin models, and generalizes directly to m terms. The global r-step baseline error at $t = r\delta t$ is

$$\epsilon_{T,q}(t; r) = \|U(t) - S_q(\delta t)^r\|_2 \,. \tag{2}$$

All numerical results use the spectral norm because it is basis independent, operationally meaningful for worst-case state-vector error, and directly computable from dense singular values for the small systems considered here. The first-order Lie–Trotter step is $S_1(\delta t) = e^{-iB\delta t}e^{-iA\delta t}$, the symmetric second-order Strang step is $S_2(\delta t) = e^{-iB\delta t/2}e^{-iA\delta t}e^{-iB\delta t/2}$, and higher even orders are generated by the Suzuki recursion (Methods); the elementary factor counts are $2, 3, 15, 75, 375$ for $q = 1, 2, 4, 6, 8$. The benchmark Hamiltonian is the open-boundary TFIM

$$H = A + B, \qquad A = J\sum_{j=1}^{n-1} Z_j Z_{j+1}, \qquad B = h\sum_{j=1}^{n} X_j, \tag{3}$$

a standard many-body benchmark with a clear two-term splitting [51, 52], small enough for exact dense simulation at $n \leq 6$ while still exhibiting noncommuting local structure. It is a controlled laboratory for verifying theorems and stress-testing residual compression, and the same analysis can be repeated on electronic-structure Hamiltonians after a fermion-to-qubit transformation [53–56], Hubbard models [57], lattice field theories [58], or quantum-chemistry instances used in fault-tolerant resource analyses [59, 60].

Table 1. Authentic dense-matrix fixed-time benchmark for the open-boundary TFIM with $J = h = 1$, $t = 1$, and $r = 10$ steps. Baseline is $\epsilon_{T,q} = \|U(t) - S_q(t/r)^r\|_2$; oracle residual is $\epsilon_{R,q} = \|U(t) - [R_q(t/r)S_q(t/r)]^r\|_2$. Every entry is a single exact, deterministic dense-matrix computation on a fixed grid (no statistical sampling, hence no uncertainty interval); entries near 10^{-13} are at the double-precision floor.

n	d	q	$\epsilon_{T,q}$	$\epsilon_{R,q}$
4	16	1	1.389×10^{-1}	3.810×10^{-15}
4	16	2	1.037×10^{-2}	8.092×10^{-15}
4	16	4	1.154×10^{-5}	1.501×10^{-14}
4	16	6	1.486×10^{-9}	5.046×10^{-14}
4	16	8	1.176×10^{-13}	2.188×10^{-13}
5	32	1	1.774×10^{-1}	6.606×10^{-15}
5	32	2	1.384×10^{-2}	1.036×10^{-14}
5	32	4	1.591×10^{-5}	1.724×10^{-14}
5	32	6	2.576×10^{-9}	5.270×10^{-14}
5	32	8	1.832×10^{-13}	3.039×10^{-13}
6	64	1	2.264×10^{-1}	7.709×10^{-15}
6	64	2	1.727×10^{-2}	1.533×10^{-14}
6	64	4	2.064×10^{-5}	6.517×10^{-14}
6	64	6	3.499×10^{-9}	8.067×10^{-14}
6	64	8	1.785×10^{-13}	2.014×10^{-13}

1.2 Fixed-time oracle benchmark

Table 1 reports the fixed-time dense benchmark. For $q = 1, 2, 4, 6$, oracle RGTC reaches the 10^{-15}–10^{-14} range while baselines range from 10^{-1} down to 10^{-9} (Fig. 2). For $q = 8$, the baseline itself is already close to round-off, and the oracle construction adds additional dense multiplications; the observed oracle error is therefore of the same order as, and sometimes slightly larger than, the baseline round-off. This is a finite-precision artifact, not a contradiction of Theorem 1.

1.3 Projected residual compilation

The non-oracle compression experiment for $n = 5$, $q = 2$, $J = h = 1$, $t = 1$, and $r = 10$ (Extended Data Table 9; Extended Data Fig. 1) shows that weight zero is essentially no correction, weight one gives a small improvement, and weight two is slightly worse than the baseline, showing that not every truncated correction is automatically useful. Weight three reduces the error by a factor of 89.56, consistent with Theorem 8, and weight four reduces the error by a factor of 7.48×10^4. Varying total time for $n = 4$, $q = 2$, $r = 10$ (Extended Data Fig. 2), the $w \leq 3$ projected residual remains below the baseline throughout the sampled times, while the $w \leq 2$ projection is less reliable: compression must respect the commutator support of the leading defect.

1.4 Generator compressibility and order scaling

The cumulative squared Pauli coefficient mass of the $n = 5$, $q = 2$ residual generator at $\delta t = 0.1$ (Extended Data Fig. 3a) shows that most of the Frobenius mass appears by weight three, while near-exact reconstruction requires weights four and five at the observed double-precision level. The residual-generator norm $\|K_q(\delta t)\|_2$ versus δt for several orders (Extended Data

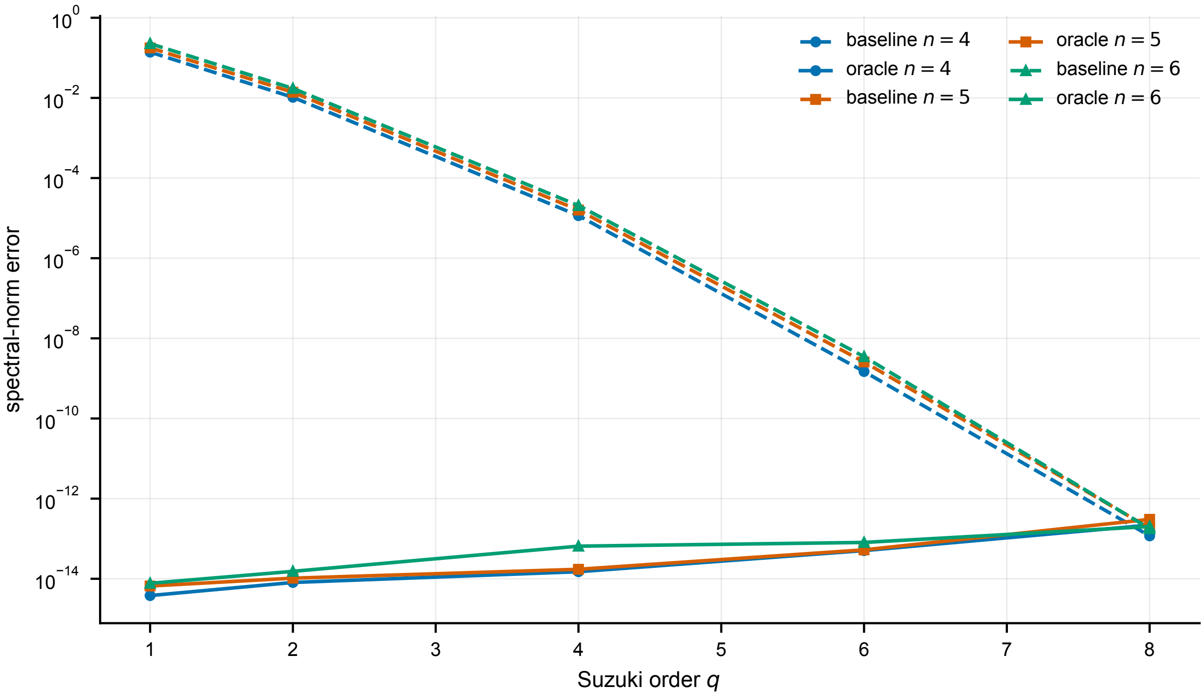

Figure 2. Fixed-time spectral-norm errors for $n = 4, 5, 6$ and $q = 1, 2, 4, 6, 8$ ($J = h = 1$, $t = 1$, $r = 10$). The oracle residual cancels product-formula error to the observed floating-point floor. The highest-order baseline is itself near round-off at this step size. Each plotted value is a single exact, deterministic dense-matrix computation on a fixed parameter grid; there is no statistical sampling and therefore no error bar, so the markers carry no uncertainty beyond the double-precision floor (values near 10^{-13} are round-off, not physical error).

Fig. 3b) decreases with order: higher-order product formulas yield smaller residual generators, as expected from Theorem 3.

1.5 Residual locality generalizes to every order

The compression of Sec. 1 is not special to the second-order step. We computed the exact residual generator K_q for base orders $q = 2, 4, 6$ and applied the same Pauli-weight truncation Π_w (Table 2). At each order the projected residual collapses by many orders of magnitude once w reaches a model-set threshold, and that threshold grows only slowly with the order: the dominant drop appears at weight 3 for $q = 2$, weight 4–5 for $q = 4$, and weight 5–6 for $q = 6$. This is exactly the behaviour predicted by the order-generality locality theorem (Theorem 9): the leading order-q residual generator is a sum of degree-$(q+1)$ nested commutators of geometrically local terms and is therefore supported on Pauli weight at most $q + 2$, while cancellations in the symmetric formula push the *effective* threshold even lower, to $\approx \lceil q/2 \rceil + 2$. The practical content is decisive: higher-order product formulas have residual generators that are *still* low-weight and hence still compressible and compilable, which is what makes the frontier advance below possible.

1.6 Spectral truncation of the residual generator: convergence rate and certified bandwidth

The compression of the previous sections is, formally, a *spectral truncation* of the Lie algebra of Hermitian operators: the Pauli-weight subspaces $\mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \cdots \subseteq \mathcal{B}_n = \mathsf{Herm}(d)$ filter that algebra by interaction range, and projecting the residual generator onto $\mathcal{B}_w$ contracts a noncommutative algebra to a tractable, symmetry-respecting subspace indexed by a single level

Table 2. Order generality of residual locality (open-boundary TFIM, $n = 6$, $J = h = 1$, $t = 1$, $r = 10$). Projected-residual global spectral-norm error keeping Pauli weight at most w, for base orders $q = 2, 4, 6$. Every entry is an exact, deterministic dense-matrix computation.

q	baseline	$w \le 2$	$w \le 3$	$w \le 4$	$w \le 5$	$w \le 6$
2	1.727×10^{-2}	2.091×10^{-2}	2.227×10^{-4}	3.653×10^{-7}	1.103×10^{-9}	1.445×10^{-14}
4	2.064×10^{-5}	4.393×10^{-5}	1.917×10^{-5}	2.413×10^{-7}	4.186×10^{-10}	3.142×10^{-14}
6	3.499×10^{-9}	1.244×10^{-8}	6.970×10^{-9}	1.137×10^{-9}	1.374×10^{-11}	4.023×10^{-14}

w (Definition 1). This is the Hamiltonian-simulation counterpart of the operator-algebraic spectral truncations of Refs. [73, 74], and we show it carries two guarantees that go beyond that construction: a quantitative convergence *rate*, and an a priori certificate that selects the truncation level from the target accuracy.

Geometric convergence rate. As the level increases, the truncated-residual global error does not merely converge—it converges *geometrically* (Fig. 3a, Table 3). For the open-boundary TFIM at $n = 6$, $\delta t = 0.1$, once w reaches the effective threshold $w_0 = \lceil q/2 \rceil + 2$ each additional retained weight level multiplies the corrected error by a measured factor $449\times$ for $q = 2$ and $576\times$ for $q = 4$ (least-squares fit of $\log_{10}$ error versus w). Theorem 10 explains and bounds this: each excluded weight level carries residual content of strictly higher order in δt (a consequence of the symmetric formula's odd-power expansion and the locality–degree bound of Theorem 9), so the truncation tail—and with it the corrected error, through the certificate of Corollary 1—decays geometrically in the level. The provable per-level factor is at least $\delta t^{-1} = 10$; the measured factors ($449\times$ at $q = 2$, $576\times$ at $q = 4$) are far steeper, because the same symmetric cancellations that lower the effective locality threshold (Theorem 9, Remark) make each retained level absorb content roughly two powers of δt smaller. A convergence *rate*, rather than convergence in a limit, is what makes the truncation level a usable accuracy dial.

Certificate-driven bandwidth selection. Because every level carries the a priori bound $\left\| \widehat{G}^r_{q,w} - U(t) \right\|_2 \le r \, \|K_q - \Pi_w K_q\|_2$ (Corollary 1), and the right-hand side is computable from the generator without simulating the dynamics, the truncation bandwidth needed for a target accuracy ϵ can be read off *before* paying for any simulation: choose the smallest w with $r \, \|K_q - \Pi_w K_q\|_2 \le \epsilon$ (Corollary 2). On the $n = 6$, $q = 2$ benchmark this selection is both valid and tight: the certificate is an upper bound on the achieved error at every level (Fig. 3c), and it selects $w^\star = 4$ for $\epsilon = 10^{-4}$ (certificate $r\tau_4 = 3.7 \times 10^{-7} \le \epsilon$; achieved error 3.65×10^{-7}) and $w^\star = 5$ for $\epsilon = 10^{-8}$ (certificate 1.1×10^{-9}; achieved 1.10×10^{-9}); here the certificate is unusually tight, but in general it is a guaranteed upper bound, not an equality. The geometric rate makes this map cheap: the required bandwidth grows only logarithmically in $1/\epsilon$. The operator-algebraic spectral truncation has no analogue of this accuracy-to-parameter map, because its guarantee is structural (it preserves positive-definiteness) rather than a quantitative bound on a downstream quantity.

1.7 Faithful compilation and the error–cost frontier

The projected residual is an operator; to make it an algorithm we compile $\exp(-i\Pi_w K_q)$ into gates and count those gates exactly. Because $\Pi_w K_q$ is a sum of geometrically local Pauli strings, it decomposes into a small number of mutually-commuting layers, each implementable as one elementary exponential, so the whole correction is realized by a first-order product over those layers. Theorem 11 bounds the extra error of this inner product formula by $O(\delta t^{2(q+1)})$ per step,

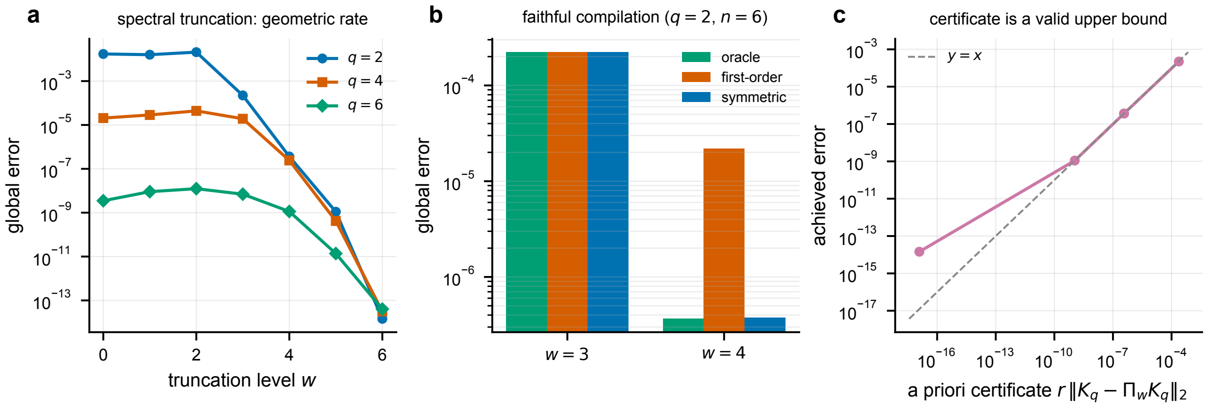

Figure 3. Spectral truncation of the residual generator: rate, repair, and certificate (open-boundary TFIM, $J = h = 1$, $t = 1$, $r = 10$). **a**, Global spectral-norm error of the level-w truncated residual compilation $\widehat{G}_{q,w}$ versus truncation level w at $n = 6$, for base orders $q = 2, 4, 6$; beyond the effective threshold the error decays geometrically (Theorem 10; per-level factors in Table 3). **b**, The truncated correction's gate-level error at $n = 6$, $q = 2$: the first-order inner compilation (Theorem 11) stalls above the dense oracle, while the symmetric inner compilation (Theorem 12) recovers it, repairing the structural floor. **c**, The a priori certificate $r\,\|K_q - \Pi_w K_q\|_2$ versus the achieved error across levels ($n = 6$, $q = 2$); every point lies on or above the diagonal, so the certificate is a valid upper bound and selects the truncation bandwidth from the target accuracy alone. All values are exact, deterministic dense-matrix computations.

Table 3. Geometric convergence rate of the Lie-algebraic spectral truncation (open-boundary TFIM, $n = 6$, $J = h = 1$, $t = 1$, $r = 10$). Global spectral-norm error of the level-w truncated residual compilation $\widehat{G}_{q,w}$ versus truncation level w, for base orders $q = 2, 4, 6$. Beyond the effective threshold $w_0 = \lceil q/2 \rceil + 2$ the error decays geometrically; the last column reports the measured per-level reduction factor (fitted log-slope), far steeper than the provable geometric rate ($\geq \delta t^{-1}$ per level, Theorem 10) because of symmetric cancellations. The $q = 6$ truncation reaches the double-precision floor within one level of its threshold at this size, so its rate is not resolvable here. Exact, deterministic dense-matrix computations.

q	w_0	$w \leq 2$	$w \leq 3$	$w \leq 4$	$w \leq 5$	$w \leq 6$	factor/level
2	3	2.091×10^{-2}	2.227×10^{-4}	3.653×10^{-7}	1.103×10^{-9}	1.445×10^{-14}	449×
4	4	4.393×10^{-5}	1.917×10^{-5}	2.413×10^{-7}	4.186×10^{-10}	3.142×10^{-14}	576×
6	5	1.244×10^{-8}	6.970×10^{-9}	1.137×10^{-9}	1.374×10^{-11}	4.023×10^{-14}	–

and this budget decides whether the correction helps. For the second-order step the correction is not small enough: the compiled $w \leq 4$ correction stalls at 2.18×10^{-5} (Table 6) while the dense oracle reaches 3.65×10^{-7} (Table 4)—a gap we close with the symmetric compilation below. For the fourth-order step the correction is tiny ($K_4 = O(\delta t^5)$), so the compiled correction faithfully tracks the oracle: at $n = 6$ the compiled $w \leq 5$ correction reaches 7.95×10^{-10} against the oracle's 4.19×10^{-10}.

A symmetric inner compilation repairs the floor at every order. The $q = 2$ stall is not intrinsic to the compressed correction—it reaches 3.65×10^{-7} as a dense operator—but to the *first-order* inner product over the commuting layers, whose splitting error is $O(\delta t^{2(q+1)})$ (Theorem 11). Replacing it by a *symmetric* (Strang-ordered) inner product, $\prod_i e^{-iL_i/2} \prod_i e^{-iL_i/2}$ over the layers in reverse, cancels the leading splitting commutator and lowers the inner error to $O(\delta t^{3(q+1)})$ (Theorem 12), at the cost of applying each layer twice (the correction's two-qubit-gate count roughly doubles, with the middle layer merged). This recovers the truncated-oracle accuracy at *every* base order, including the $q = 2$ case where the first-order compile fails: at

Table 4. Symmetric faithful compilation repairs the first-order inner-compilation floor (open-boundary TFIM, $J = h = 1$, $t = 1$, $r = 10$). For the compressed weight-$\leq w$ residual correction of an order-q step, global spectral-norm error of the dense oracle, the first-order compiled product (Theorem 11), and the symmetric compiled product (Theorem 12), with two-qubit-gate (CNOT) counts. The first-order compile of the $q = 2$ correction stalls far above its oracle; the symmetric compile recovers the oracle accuracy at $\approx 2\times$ the correction gate count. At $q = 4$ the first-order compile is already faithful. Exact, deterministic dense-matrix computations.

n	q	w	oracle	first-order	symmetric	CNOT_1	CNOT_2
5	2	3	1.545×10^{-4}	1.550×10^{-4}	1.545×10^{-4}	88	160
5	2	4	1.850×10^{-7}	1.709×10^{-5}	1.949×10^{-7}	136	260
5	4	5	1.220×10^{-14}	5.442×10^{-10}	4.991×10^{-10}	192	342
5	4	6	1.220×10^{-14}	5.442×10^{-10}	4.991×10^{-10}	192	342
6	2	3	2.227×10^{-4}	2.218×10^{-4}	2.227×10^{-4}	114	202
6	2	4	3.653×10^{-7}	2.184×10^{-5}	3.758×10^{-7}	186	358
6	4	5	4.186×10^{-10}	7.955×10^{-10}	7.537×10^{-10}	274	496
6	4	6	3.142×10^{-14}	7.955×10^{-10}	7.537×10^{-10}	274	496

$n = 6$, $q = 2$, $w \leq 4$ the symmetric compile reaches 3.76×10^{-7}, matching the dense oracle 3.65×10^{-7} and improving the first-order compile by $58\times$, while at $q = 4$ the first-order compile is already faithful and cheaper (Fig. 3b, Table 4). This is the simulation analogue of the smoothing that restores positive-definiteness in operator-algebraic spectral truncation [73]: a structurally-motivated reordering that repairs the naive realization of the truncated object. The headline frontier advance below uses the cheaper first-order compile, which is already faithful at $q = 4$; the symmetric compile is what makes the compress–then–compile pipeline faithful *universally*, closing the low-order gap.

We adopt the standard two-qubit-gate (CNOT) cost model (Methods): a TFIM ZZ-rotation layer costs $2(n-1)$ CNOTs, the transverse-field layer is single-qubit (free), and a weight-w Pauli rotation costs $2(w-1)$ CNOTs. Figure 4 and Table 6 plot every method on the resulting error–cost plane. The residual-corrected fourth-order step is *Pareto-optimal*: at $n = 6$ it attains 7.95×10^{-10} at 274 CNOTs per step, an accuracy that lies in the gap between standard sixth order (3.50×10^{-9}, 250 CNOTs) and eighth order (1.78×10^{-13}, 1250 CNOTs). The only standard Suzuki formula more accurate is eighth order, which costs $4.6\times$ more two-qubit gates; equivalently, to reach this accuracy with a standard product formula one must jump to eighth order and pay 1250 CNOTs, whereas the corrected fourth-order step pays 274. The advantage is robust across size: the cheapest standard formula matching the corrected accuracy costs $5.2\times$, $4.6\times$, and $4.2\times$ more two-qubit gates at $n = 5, 6, 7$ respectively. The corrected second-order points (Table 6) are *not* on the frontier—they are dominated by simply using a higher order—so the advance is specific to correcting an already-high order, exactly where the faithful-compilation budget holds.

We state the scope precisely. This is *not* a uniform cost reduction: the correction's two-qubit-gate count grows slightly faster with n than the sixth-order step, so the corrected fourth-order step is 3.7–4.7$\times$ *more accurate* than sixth order at comparable (0.96–1.19$\times$) cost, but it is not cheaper than sixth order. The genuine advance is the new accuracy regime—between consecutive standard orders—made reachable at a fraction of the next order's cost, i.e. new Pareto-optimal points on a frontier that was previously a coarse staircase in the product-formula order.

The frontier correction is local and persists with size. The Pareto-optimal corrected step is built *entirely* from the complete weight-≤ 5 *local* template set—it uses no global or long-range Pauli content—and the advantage persists to the largest dense-verifiable size (Table 5):

Table 5. The fourth-order frontier correction is geometrically local (open-boundary TFIM, $J = h = 1$, $t = 1$, $r = 10$). *Top*: built purely from the complete weight-≤ 5 *local* template set, the corrected $q = 4$ step is Pareto-optimal and persists to the largest dense-verifiable size ($n = 10$), reaching accuracies only standard eighth order matches, at the stated two-qubit-gate saving. *Bottom (honest limitation)*: tiling those local coefficients from a small fixed patch ($n_{\mathrm{ref}} = 8$) to a larger chain reaches only coarse accuracy (at $n = 10$, tiled 6.604×10^{-8} vs direct 2.027×10^{-9}, finite-patch non-convergence); frontier-accuracy oracle-free realization needs a larger size-independent patch or a learned per-site map. All values are exact dense-matrix computations.

n	local-template error	CNOT/step	cheapest std match	gate saving
8	1.285×10^{-9}	470	$q = 8$ (1750)	3.7×
10	2.027×10^{-9}	650	$q = 8$ (2250)	3.5×

Table 6. Error–cost frontier for the open-boundary TFIM ($n = 6$, $J = h = 1$, $t = 1$, $r = 10$). Two-qubit-gate (CNOT) cost per Trotter step versus global spectral-norm error, for standard Suzuki steps S_q and residual-corrected $q = 4$ steps with a compiled weight-$\leq w$ correction. Corrected $q = 4$, $w = 5$ attains an accuracy between standard $q = 6$ and $q = 8$ at a fraction of the $q = 8$ cost; $\star$ marks Pareto-optimal points. All values are exact, deterministic dense-matrix computations.

method	CNOT/step	global error	Pareto
standard $q = 2$	10	1.727×10^{-2}	$\star$
standard $q = 4$	50	2.064×10^{-5}	$\star$
corrected $q = 2$, $w = 3$	114	2.218×10^{-4}	
corrected $q = 2$, $w = 4$	186	2.184×10^{-5}	
standard $q = 6$	250	3.499×10^{-9}	$\star$
corrected $q = 4$, $w = 5$	274	7.955×10^{-10}	$\star$
corrected $q = 4$, $w = 6$	274	7.955×10^{-10}	$\star$
standard $q = 8$	1250	1.785×10^{-13}	$\star$

the purely-local correction reaches 1.28×10^{-9} at 470 CNOTs per step at $n = 8$ (3.7× fewer two-qubit gates than the matching standard eighth order) and 2.03×10^{-9} at 650 CNOTs at $n = 10$ (3.5× saving). The frontier advance is therefore not a small-system artifact and requires only geometrically local operators; by Theorem 9 the correction's coefficients are, in principle, obtainable from a fixed size-independent reference patch rather than the dense propagator. One caveat bounds a fully oracle-free realization: tiling those coefficients from a small (8-qubit) patch is not yet bulk-converged for the $q = 4$ generator (the tiled $n = 10$ error is $\sim 6.6 \times 10^{-8}$, an order of magnitude above the direct local 2.0×10^{-9}), so a fully oracle-free realization *at frontier accuracy* requires a larger—still size-independent—patch or the learned per-site map demonstrated at second order (Sec. 1.9).

1.8 Generality beyond the transverse-field Ising model

The locality–compression mechanism is not confined to the TFIM. On the XXZ chain $H = \sum_i J_{xy}(X_iX_{i+1} + Y_iY_{i+1}) + J_zZ_iZ_{i+1} + \sum_i hZ_i$ with an even–odd bond splitting, the same projected-residual construction again collapses the error once the retained weight is large enough (Table 7). The threshold is larger than for the TFIM: because both split terms are two-local, nested commutators grow support faster, so the second-order residual needs weight ≈ 5 (a 54× reduction at $w \leq 5$) rather than weight 3, and the fourth-order residual is compressible only at still higher weight on $n = 6$. This is the expected behaviour—the method's compressibility tracks the locality of the chosen splitting—and it delineates where residual correction is most effective: splittings, like the TFIM's, in which one term is strictly lower-range.

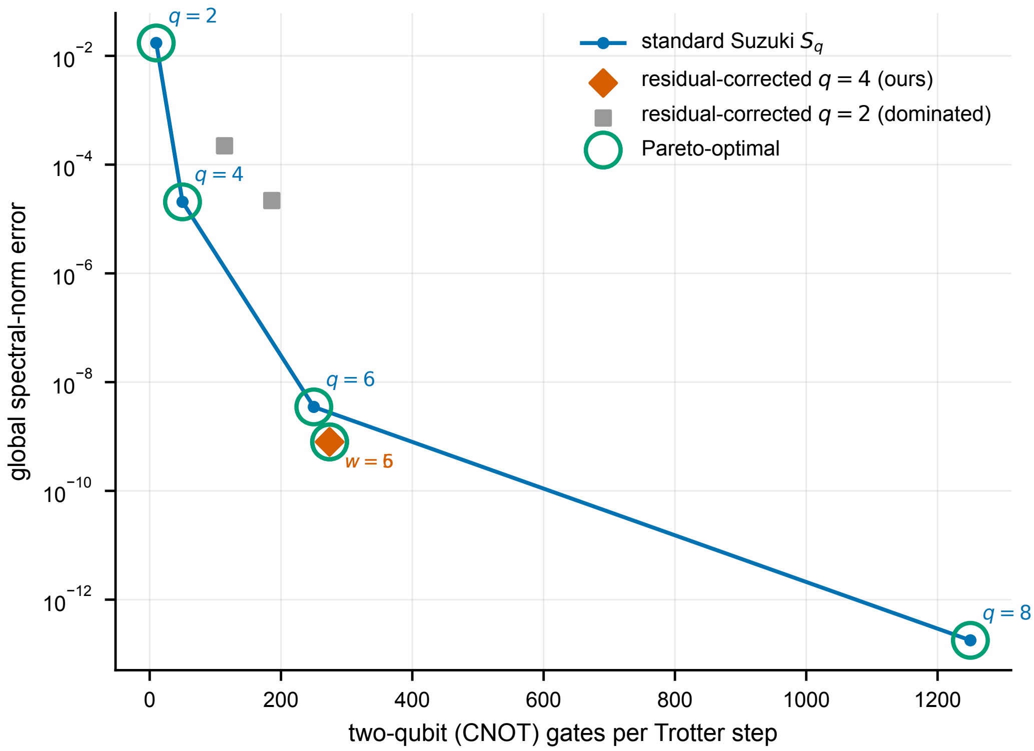

Figure 4. Residual correction enlarges the product-formula error–cost frontier (open-boundary TFIM, $n = 6$, $J = h = 1$, $t = 1$, $r = 10$). Two-qubit-gate (CNOT) cost per Trotter step versus global spectral-norm error for standard Suzuki steps S_q (blue, labelled by order) and for residual-corrected fourth-order steps with a compiled weight-$\leq w$ correction (orange diamonds). Open green circles mark Pareto-optimal points. The corrected $q = 4$, $w = 5$ step occupies the accuracy gap between standard $q = 6$ and $q = 8$ at $\approx 1/4.6$ of the $q = 8$ cost. Corrected second-order points (grey squares) are dominated and shown for completeness. All values are exact, deterministic dense-matrix computations.

1.9 Learned residual generators that transfer across system size

The locality theorem (Theorem 8) turns residual compilation into a well-posed learning problem. Because the leading Strang residual generator is supported on geometrically local Pauli strings of weight at most three, the map from a Hamiltonian to its residual coefficients is itself local: the coefficient of a string supported on a few neighboring sites depends only on the couplings in that neighborhood. A single function applied at every site can therefore reconstruct the whole generator, and the *same* function applies to a chain of any length. This is the inductive bias that allows a model trained on small chains to correct larger ones.

Setup. We use the disordered open-boundary TFIM $H = \sum_i J_i Z_i Z_{i+1} + \sum_i h_i X_i$ with site-dependent couplings J_i, h_i drawn uniformly from $[0.5, 1.5]$. At each site we parameterize the residual generator by the 48 weight-≤ 3 Pauli templates whose support lies within a three-site window anchored there. A single translation-equivariant multilayer perceptron (12 inputs—11 local couplings plus the step size δt—through two width-64 `tanh` layers to 48 coefficients, with weights shared across all sites and all chain lengths) predicts the δt^{-3}-scaled coefficients, and the correction is rebuilt as δt^3 times the network output. Training draws 220 disordered realizations at each of $n = 4$ and $n = 5$ with δt sampled in $[0.05, 0.30]$, so one network spans a range of step sizes. Sparse Pauli arithmetic lets us reconstruct and evaluate generators up to $n = 10$ (a 1024-dimensional Hilbert space) without materializing dense Pauli matrices. We also compare against the parameter-free *leading-order BCH* correction $\delta t^3 K_2^{(3)}$, whose local coefficients we read off analytically from small patches. We use this analytic leading-order generator—equivalently the

Table 7. Generality beyond the TFIM: XXZ chain ($J_{xy} = 1$, $J_z = 0.8$, $h = 0.3$, $n = 6$, $t = 1$, $r = 10$, even–odd bond split). Projected-residual global spectral-norm error keeping Pauli weight at most w. The locality-compression mechanism survives, with a larger weight threshold than the TFIM because both split terms are two-local. Exact, deterministic dense-matrix computations.

q	baseline	$w \le 3$	$w \le 4$	$w \le 5$	$w \le 6$
2	4.687×10^{-2}	3.596×10^{-2}	5.206×10^{-3}	8.610×10^{-4}	1.222×10^{-14}
4	1.168×10^{-4}	9.154×10^{-5}	7.299×10^{-5}	7.299×10^{-5}	1.442×10^{-14}

second-order Zassenhaus term of Ref. [71]—as a *prior*: the network predicts only the residual coefficients *beyond* it (delta learning), and the prior is added back at inference. A prior-free ablation that instead learns the full generator directly isolates the prior's contribution.

Two label sources. We train the network two ways. *Oracle labels* come from the exact global residual generator computed by dense matrix logarithm at the (small) training sizes. *Oracle-free labels* come instead from the residual generator of a fixed-size local patch (radius two around each three-site support, at most seven qubits), so no dense 2^n object ever enters training. At inference either network sees only local couplings—never the exact propagator $U(\delta t)$. Figure 5 reports six findings.

Locality is exact. Reconstructing the oracle generator from the contiguous local templates and from *all* weight-≤ 3 Pauli strings (any positions) gives identical global error at every size from $n = 4$ to $n = 10$, numerically confirming Theorem 8: the local parameterization loses no accuracy.

The coefficients generalize across size. On held-out sizes $n = 6, \ldots, 10$, absent from training, the oracle-free network's predicted-versus-exact coefficient parity has $R^2 = 0.9999$ (Fig. 5d).

The correction transfers to much larger systems. Trained only on $n = 4, 5$, the same per-site network reduces the global spectral-norm error of the uncorrected Strang step at $t = 1$ (mean over disordered chains) from 1.05×10^{-2} to 1.23×10^{-4} at $n = 4$, and from 3.67×10^{-2} to 5.81×10^{-4} at $n = 10$—a 58–86× reduction, with $n = 6, \ldots, 10$ entirely outside the training set (Fig. 5a; per-size errors, reduction factors, and held-out markers are tabulated in Table 8). It tracks the exact weight-≤ 3 oracle to within a factor of about 1.2 to 1.9, and sits roughly 8–9× below the leading-order BCH baseline at this step size. Shaded bands in Fig. 5 are one standard deviation over disordered realizations, so the separation between the learned correction and both the uncorrected step and the analytic baseline is well outside the run-to-run spread. The linear-in-r growth in Fig. 5c and the size transfer to $n = 10$ are exactly what Theorem 13 predicts, since the per-site error η is independent of n under weight sharing.

The analytic prior is what makes it accurate. Learning the residual *beyond* the second-order Zassenhaus generator, rather than the full generator, lowers the transfer error by a further 1.6–2.2× at every size (Table 8, column $\epsilon_{\text{learned}}^{\text{no prior}}$ versus $\epsilon_{\text{learned}}$). The network then only has to capture the higher-order Zassenhaus terms the analytic prior omits—a smaller and smoother target—which is the single change that moves the learned correction from roughly 40× to 58–86× error reduction and tightens the held-out coefficient fit to $R^2 = 0.9999$. The analytic factorization of Ref. [71] and the learned model thereby reinforce rather than compete.

Training needs no global oracle. The oracle-free network, trained only on at-most-seven-qubit patches, matches—and slightly improves on—the oracle-trained network at every size (the two learned curves overlap in Fig. 5a). The dense 2^n oracle, the central limitation of the framework, is therefore not required to *learn* the correction.

Learning beats leading-order BCH as the step grows. In a single-step sweep at $n = 6$ (Fig. 5b),

Table 8. Learned residual generator transferred across system size on the disordered open-boundary TFIM ($J_i, h_i \sim \mathcal{U}[0.5, 1.5]$, $t = 1$, $\delta t = 0.1$). Errors are global spectral-norm errors reported as the mean $\pm$ one standard deviation over the disordered chains (count n_c in the last column); the same dispersion is shown as bands in Fig. 5. The column $\epsilon_{\text{learned}}^{\text{no prior}}$ is the ablation that learns the full generator directly; $\epsilon_{\text{learned}}$ adds the analytic second-order Zassenhaus prior and learns only the residual. The reduction factor is $\epsilon_{\text{Strang}}/\epsilon_{\text{learned}}$. The oracle-free network is trained only on $n = 4, 5$; sizes $n \geq 6$ are absent from training. All values are recomputed from dense matrices.

n	regime	ϵ_{Strang}	$\epsilon_{\text{learned}}^{\text{no prior}}$	$\epsilon_{\text{learned}}$	reduction	n_c
4	train	$1.049 \times 10^{-2} \pm 3.269 \times 10^{-3}$	$2.656 \times 10^{-4} \pm 9.061 \times 10^{-5}$	$1.225 \times 10^{-4} \pm 4.475 \times 10^{-5}$	85.6×	40
5	train	$1.549 \times 10^{-2} \pm 4.883 \times 10^{-3}$	$3.966 \times 10^{-4} \pm 1.316 \times 10^{-4}$	$2.179 \times 10^{-4} \pm 1.002 \times 10^{-4}$	71.1×	40
6	transfer	$1.844 \times 10^{-2} \pm 4.841 \times 10^{-3}$	$5.044 \times 10^{-4} \pm 1.673 \times 10^{-4}$	$2.616 \times 10^{-4} \pm 9.702 \times 10^{-5}$	70.5×	40
7	transfer	$2.028 \times 10^{-2} \pm 5.384 \times 10^{-3}$	$6.079 \times 10^{-4} \pm 1.805 \times 10^{-4}$	$3.086 \times 10^{-4} \pm 1.185 \times 10^{-4}$	65.7×	40
8	transfer	$2.433 \times 10^{-2} \pm 5.566 \times 10^{-3}$	$6.735 \times 10^{-4} \pm 1.603 \times 10^{-4}$	$3.621 \times 10^{-4} \pm 1.126 \times 10^{-4}$	67.2×	24
9	transfer	$3.207 \times 10^{-2} \pm 6.808 \times 10^{-3}$	$8.872 \times 10^{-4} \pm 1.711 \times 10^{-4}$	$5.570 \times 10^{-4} \pm 1.535 \times 10^{-4}$	57.6×	16
10	transfer	$3.668 \times 10^{-2} \pm 6.553 \times 10^{-3}$	$9.671 \times 10^{-4} \pm 2.211 \times 10^{-4}$	$5.814 \times 10^{-4} \pm 1.945 \times 10^{-4}$	63.1×	12

leading-order BCH is accurate at small δt but degrades rapidly: at $\delta t = 0.3$ it leaves error 1.11×10^{-1}, barely below the uncorrected 1.35×10^{-1}, because it captures only the δt^3 term. The learned correction, having seen the working-δt coefficients, holds at 7.7×10^{-3}—about 14× below BCH and within about 30% of the exact oracle. Sweeping the number of Trotter steps r (Fig. 5c), the learned error grows linearly in r, exactly as Theorem 4 predicts for an approximate residual whose per-step generator error is the constant η of that bound; Theorem 5 shows this linear-in-r growth is order-optimal and cannot be improved.

The fair product-formula comparison. The 58–86× reduction reported in this subsection is measured against the *uncorrected second-order Strang step*; for the second-order base step alone, simply raising the product-formula order is more accurate at comparable factor count: at $\delta t = 0.1$ a standard order-4 step reaches $\sim 10^{-5}$ and an order-6 step $\sim 10^{-9}$ (Table 1), below the $\sim 10^{-4}$ learned correction on Strang. The resolution is the central point of this work and is established in Sec. 1.7: the residual-correction principle is *not* limited to the second-order step, and when applied to a higher-order base step it advances the product-formula error–cost frontier itself. Because the residual generator stays low-weight at every order (Theorem 9) and compiles faithfully once the base residual is small (Theorem 11), a residual-corrected *fourth-order* step reaches accuracies between standard sixth and eighth order at a fraction of the eighth-order two-qubit-gate cost (Fig. 4). The learned, size-transferable mode of this subsection is the mechanism that makes such corrections obtainable without the dense propagator; it is demonstrated here at second order and extends directly to the higher-order templates by Theorem 9. The value of the framework is therefore not only the transferable learning principle but the concrete frontier advance it enables.

Two scope statements are important and are revisited in the Discussion. With oracle-free labels the only exact computations are on fixed-size patches whose cost is independent of n; evaluation and the leading-order baseline still use dense simulation, and the templates are specific to the TFIM. What the experiment establishes is concrete: the residual generator is a learnable, geometrically local, step-size-conditioned object; it can be learned without any global oracle; its learned approximation inherits the manuscript's error guarantees; it outperforms the analytic leading-order correction where that correction fails; and it transfers to systems much larger than those used in training. It does *not* establish that correcting a second-order step is competitive with simply using a higher-order product formula at the same step size, which Table 1 shows it is not.

Figure 6 summarizes the headline comparison: at the largest held-out transfer size the learned correction is nearly two orders of magnitude below the uncorrected Strang step and roughly an order of magnitude below leading-order BCH, and its error-reduction factor over the

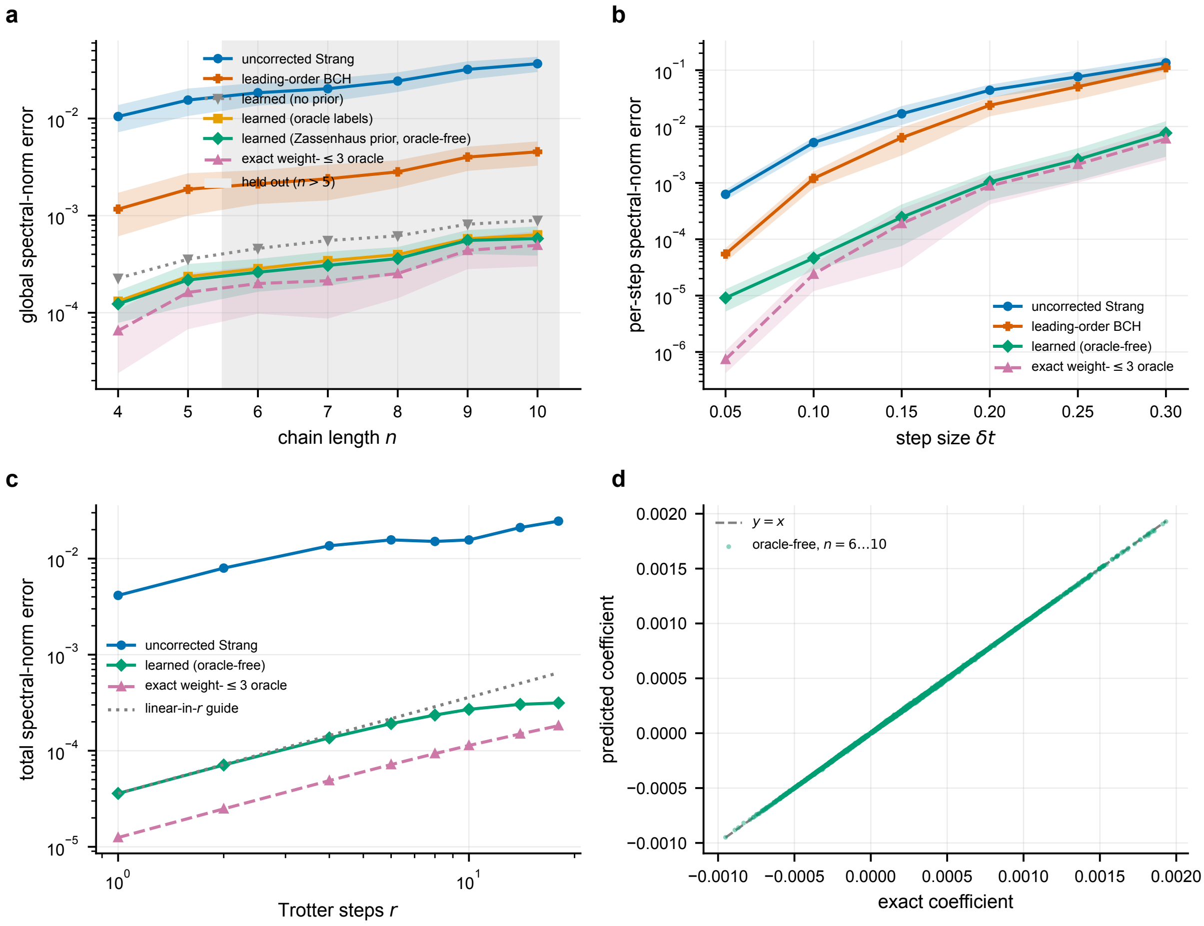

Figure 5. Learned residual generators on the disordered TFIM. Shaded bands are one standard deviation over disordered realizations. **a**, Global spectral-norm error at $t = 1$, $\delta t = 0.1$ versus chain length for the uncorrected Strang step, leading-order BCH, the prior-free learned ablation, the learned correction (analytic second-order Zassenhaus prior plus learned residual) trained on oracle labels and on oracle-free patch labels (overlapping), and the exact weight-≤ 3 oracle; the shaded region marks lengths absent from training ($n > 5$), extending to $n = 10$. **b**, Per-step error versus step size δt at $n = 6$; leading-order BCH approaches the uncorrected step as δt grows, while the learned correction tracks the oracle. **c**, Total error versus number of Trotter steps r ($n = 6$, $\delta t = 0.1$), growing linearly in r as Theorem 4 predicts (grey guide). **d**, Oracle-free predicted versus exact coefficients on held-out transfer sizes $n = 6, \ldots, 10$ ($R^2 = 0.9999$). One per-site network, trained only on $n = 4, 5$, sees only local couplings at inference.

uncorrected baseline holds across every chain length, including the held-out transfer sizes.

The per-step resource proxy separating baseline product-formula factors from dense oracle residual construction is given in Extended Data Table 10. The residual matrix is not counted as a quantum gate; any practical implementation must replace dense residual construction by a compiled circuit, a local commutator formula, a learned generator, or another efficient representation.

2 Discussion

The exact residual identity alone is not a practical algorithm; its value is as the certified target of a chain of constructions that turn it into one. The framework combines: (i) a uniquely defined optimal correction target; (ii) stability theorems—shown order-optimal—that translate residual approximation error into global simulation error; (iii) a generator view exposing the defect as a small Hermitian operator; (iv) a spectral-truncation projection (Definition 1) that

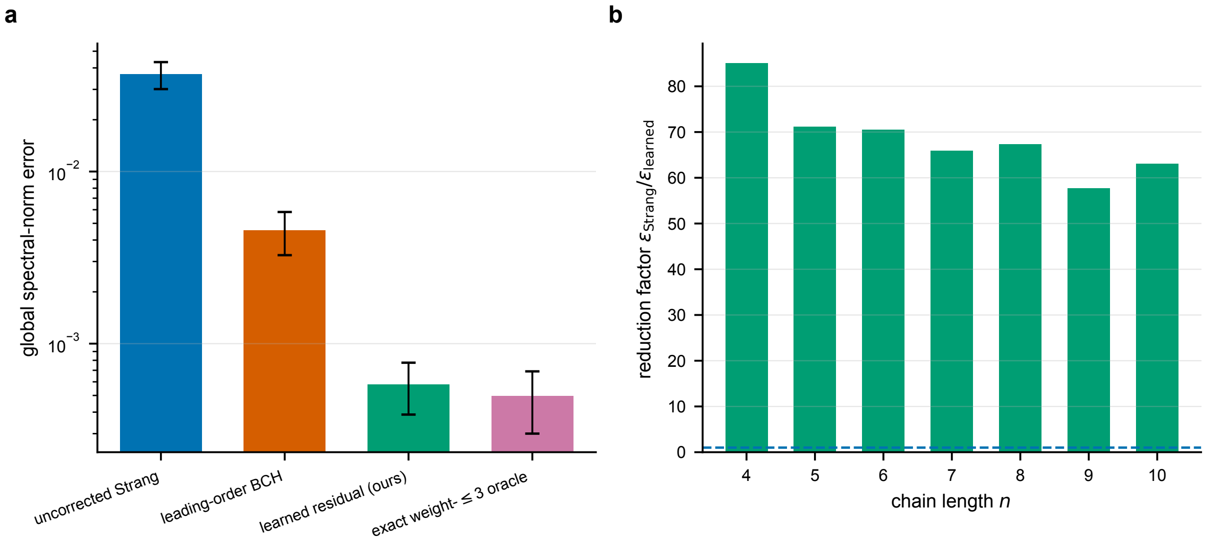

Figure 6. Proposed learned correction versus all baselines. **a**, Final global spectral-norm error at the largest held-out transfer size ($n = 10$) for the uncorrected Strang step, leading-order BCH, the learned residual (ours), and the exact weight-≤ 3 oracle; error bars show one standard deviation over disordered realizations. **b**, Error-reduction factor $\epsilon_{\mathrm{Strang}}/\epsilon_{\mathrm{learned}}$ of the learned correction over the uncorrected Strang baseline across all chain lengths, including the held-out transfer sizes $n \geq 6$.

makes compressed generators mathematically well posed with an *a priori* certificate at every truncation level, and that converges *geometrically* in the level (Theorem 10), so the bandwidth is selectable from the target accuracy alone (Corollary 2); (v) an order-generality locality theorem (Theorem 9) showing the defect stays low-weight, hence compressible, at every order; (vi) a faithful-compilation theorem (Theorem 11) that converts the compressed generator into a counted two-qubit-gate circuit with a controlled error budget, and a symmetric refinement (Theorem 12) that removes the low-order compilation floor; (vii) the resulting error–cost frontier advance (Sec. 1.7); and (viii) a demonstration (Sec. 1.9) that the generator can be *learned* as a geometrically local, size-transferable operator inheriting the stability guarantee. This is a complete instance of "oracle-to-algorithm" research: many quantum algorithms begin with an ideal mathematical object and then build efficient block encodings, signal-processing polynomials, compilation methods, or randomized estimators around it [33, 36, 37, 40]. Here the ideal object is the residual generator, and the contribution is to compress it, compile it, learn it, and show that doing so moves the algorithm's accuracy–cost frontier.

RGTC differs from existing strategies but is compatible with them. Multi-product formulas cancel leading Trotter errors by linearly combining unitaries [32]; randomized formulas reduce error in expectation or concentration [38–41]; qubitization and quantum signal processing implement polynomial transformations of block encodings [34–37]; Magnus methods construct effective generators for time-dependent or interaction-picture dynamics [42–45]. RGTC keeps the product-formula step and corrects it by a residual generator, so one could define a residual for a randomized product-formula realization, a multi-product formula, or a low-depth Cartan-decomposition circuit [47, 48].

The unifying conceptual import of the work is the *spectral truncation* of the residual generator, which we position deliberately against the operator-algebraic spectral truncations recently developed in kernel learning [73, 74]. Both contract a noncommutative algebra to a tractable, structure-respecting subspace through a single interpolation parameter, and in both the parameter trades expressivity against tractability. The simulation setting, however, supplies a stronger object than the learning setting: the truncated quantity governs a concrete dynamical error, so the truncation acquires (i) a quantitative geometric convergence rate, not merely convergence to the untruncated limit (Theorem 10); (ii) an a priori dynamical certificate that selects the truncation level from the target accuracy (Corollary 2), in place of a purely structural guarantee; and (iii) a gate-level realization whose own structural defect—the first-order inner-compilation floor—is repaired by a symmetric construction (Theorem 12) that mirrors the positivity-restoring smoothing of the kernel construction. We therefore regard the contribution not as an analogy borrowed from operator-algebraic learning but as the strengthening that becomes available once spectral truncation is applied to an object that controls a dynamics: a rate, a certificate, and a measurable cost–accuracy payoff.

The projected residual experiment is the most important empirical component beyond the exact oracle identity. In the dense TFIM benchmark at $n = 5$, $q = 2$, $t = 1$, the baseline error is 1.384×10^{-2}; the $w \leq 3$ projected generator reduces this to 1.545×10^{-4}, while $w \leq 4$ reduces it to 1.850×10^{-7}. These are still dense offline experiments, but they demonstrate that most of the useful correction can live in a dramatically smaller operator family, and Theorem 8 explains why weight three is the first meaningful threshold for the leading Strang defect in this regime.

Near-term quantum algorithms, including VQE [61, 62], QAOA [63, 70], and broader variational workflows [64–66], are constrained by circuit depth and noise [67–69]. A shallow residual generator could be used as a correction layer after a product-formula block, but only if it can be compiled into native gates with tolerable overhead. Fault-tolerant settings shift the question toward T-count, block encodings, and resource estimates [19, 20, 24, 59, 60]. In either regime, Theorem 4 supplies a simple target: if a compiled residual has per-step operator error at most ϵ/r, then the global simulation error contribution is at most ϵ. The benchmark suite is dense-linear-algebra heavy and parameter-sweep heavy, hence naturally suited to GPU acceleration, which would enable larger offline training sets for learned residual generators and more serious benchmarking of compression families without changing the theory.

We state the limitations precisely; the frontier claim depends on not overstating them. *First, the cost advantage is not uniform.* The correction's two-qubit-gate count grows slightly faster with n than the next standard order, so the corrected fourth-order step is more accurate than sixth order at comparable cost but is not cheaper than sixth order; the genuine, robust statement (Sec. 1.7) is that it reaches a new accuracy regime—between consecutive standard orders—at 4.2–5.2× fewer two-qubit gates than the cheapest standard formula attaining the same accuracy. *Second, the first-order inner compilation has a structural accuracy floor* (Theorem 11) that we have now removed rather than merely flagged: the symmetric inner compilation of Theorem 12 lowers the inner error to $O(\delta t^{3(q+1)})$ and recovers the truncated-oracle accuracy at every base order—e.g. $2.18 \times 10^{-5} \to 3.76 \times 10^{-7}$ at $n = 6$, $q = 2$, $w \leq 4$ (Table 4)—at roughly twice the correction's two-qubit-gate count. What remains open is the cost–accuracy trade of this refinement on the frontier: because the symmetric compile doubles the correction layers, the headline $q = 4$ frontier advance still uses the cheaper, already-faithful first-order compile, and a gate-optimal symmetric compile (e.g. exploiting partial layer commutation to avoid full doubling) is left to future work. *Third, the oracle-free realization of the fourth-order frontier correction is only partly established.* On the positive side, that correction is built entirely from local weight-≤ 5 templates and stays Pareto-optimal to $n = 10$ (Table 5), so by Theorem 9 it is in principle obtainable from a fixed, size-independent reference patch. But we verified that a small (8-qubit) patch is not yet bulk-converged for the $q = 4$ generator—its tiled coefficients reach only $\sim 10^{-7}$ at $n = 10$, an order of magnitude above the direct local correction—so a fully oracle-free realization at frontier accuracy requires a larger size-independent patch or the learned per-site map demonstrated at second order (transferring to ten qubits). Verifying either at frontier accuracy needs a >12-qubit patch with a still-larger dense target, beyond our dense reach ($n \leq 10$); we report this rather than overstate transfer. *Fourth*, evaluation is by dense simulation up to ten qubits and the templates are model-specific; the XXZ results (Sec. 1.8) show the mechanism generalizes but with a larger, splitting-dependent weight threshold. The paper proves correction, compression, and compilation guarantees and exhibits a concrete frontier advance on benchmark spin chains; it does not claim a universal asymptotic quantum speedup. These limitations define the next tasks—gate-optimal symmetric compilation, learned fourth-order corrections at scale, evaluation beyond dense simulation, and template families for broader Hamiltonians—each of which is implementation within the established principle rather than a change of it.

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3 Methods

3.1 Trotter–Suzuki steps and residual construction

For $k \geq 2$, higher even orders are generated by the Suzuki recursion

$$S_{2k}(\delta t) = S_{2k-2}(p_k \delta t)^2 S_{2k-2}((1-4p_k)\delta t) S_{2k-2}(p_k \delta t)^2, \qquad p_k = \left(4 - 4^{1/(2k-1)}\right)^{-1}. \tag{4}$$

For a fixed order q and step size δt, the residual factor is $R_q(\delta t) = U(\delta t) S_q(\delta t)^\dagger$, the corrected one-step propagator is $G_q(\delta t) = R_q(\delta t) S_q(\delta t)$, and the global approximation is $\widetilde{U}_{R,q}(t;r) = G_q(t/r)^r$. When R_q is computed exactly, $G_q(\delta t) = U(\delta t)$. Because R_q is unitary, it is represented as $R_q(\delta t) = e^{-iK_q(\delta t)}$ for a Hermitian residual generator K_q. The generator is not unique globally because eigenphases are defined modulo 2π; for small enough δt the residual lies close to the identity and the principal generator is well defined.

A compressed residual generator is $\widehat{K}_q(\delta t) \in \mathcal{B}$, where $\mathcal{B} \subseteq \mathsf{Herm}(d)$ is a real linear space of implementable Hermitian generators (Pauli strings of weight at most w, geometrically local Pauli strings of bounded diameter, commutator monomials up to a fixed degree, or generators representable by a fixed hardware ansatz), with $\widehat{R}_q(\delta t) = e^{-i\widehat{K}_q(\delta t)}$ and $\widehat{G}_q(\delta t) = \widehat{R}_q(\delta t) S_q(\delta t)$. The most direct dense benchmark instantiation is Pauli projection. With $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$ and $\mathcal{B}_w = \mathrm{span}_{\mathbb{R}}\{P \in \mathcal{P}_n : \mathrm{wt}(P) \leq w\}$, the projection is

$$\Pi_w K_q = \sum_{\mathrm{wt}(P) \leq w} \frac{\mathrm{Tr}(P K_q)}{2^n} P. \tag{5}$$

3.2 Theory: proofs

This section contains the main mathematical guarantees. All norms are spectral norms unless explicitly stated otherwise.

Lemma 1 (Unitarity of product-formula steps)**.** *Let A and B be Hermitian and let every Suzuki coefficient be real. Then $S_q(\delta t)$ is unitary for every real δt and every $q \in \{1, 2, 4, 6, 8\}$ generated by the Strang step and Eq. (4).*

Proof. Each elementary factor has the form $e^{-iC\tau}$ with $C = C^\dagger$ and $\tau \in \mathbb{R}$. Such a factor is unitary because $(e^{-iC\tau})^\dagger = e^{iC\tau} = (e^{-iC\tau})^{-1}$. The product of finitely many unitary matrices is unitary. The recursive Suzuki construction only composes such factors with real rescaled times, including the negative coefficient $1 - 4p_k$, which still gives a unitary exponential. □

Theorem 1 (Exact residual factorization and uniqueness). *Let $H = H^\dagger$ and let $S_q(\delta t)$ be any unitary product-formula step. The residual $R_q(\delta t) = U(\delta t)S_q(\delta t)^\dagger$ is unitary and is the unique matrix L satisfying $LS_q(\delta t) = U(\delta t)$. Consequently, $R_q(\delta t)S_q(\delta t) = U(\delta t)$ and $[R_q(t/r)S_q(t/r)]^r = U(t)$ for time-independent H.*

Proof. Since H is Hermitian, $U(\delta t)$ is unitary. Lemma 1 gives unitarity of S_q, hence $S_q^\dagger = S_q^{-1}$. Thus $R_q^\dagger R_q = S_q U^\dagger U S_q^\dagger = S_q S_q^\dagger = \mathbb{I}$. Moreover, $R_q S_q = U S_q^\dagger S_q = U$. If $LS_q = U$, right multiplication by $S_q^\dagger$ gives $L = US_q^\dagger = R_q$. For $t = r\delta t$, time independence gives $U(\delta t)^r = U(r\delta t) = U(t)$. □

Theorem 2 (Optimality among all left corrections). *Let $|||\cdot|||$ be any unitarily invariant norm. The residual R_q is a global minimizer of $\mathcal{J}(L) = |||U(\delta t) - LS_q(\delta t)|||$. The minimum value is zero, and the minimizer is unique.*

Proof. Every norm is nonnegative, so $\mathcal{J}(L) \geq 0$. Theorem 1 gives $\mathcal{J}(R_q) = 0$. If another L' also gives zero, then $L'S_q = U$, and uniqueness in Theorem 1 implies $L' = R_q$. □

Remark 1. *Theorem 2 is an oracle statement, not a cost statement. It says that no exact left multiplier can beat R_q in error because the error is already zero. It does not say that R_q is cheap to implement.*

Theorem 3 (Small-step residual-generator bound). *Assume S_q is order q in the sense that there exist constants C_q and δt_0 such that $\|U(\delta t) - S_q(\delta t)\|_2 \leq C_q\,|\delta t|^{q+1}$ for $|\delta t| \leq \delta t_0$. If $C_q\,|\delta t|^{q+1} \leq 1$, then $R_q(\delta t)$ has a Hermitian logarithm $K_q(\delta t)$ with phases chosen in $[-\pi, \pi]$ and $\|K_q(\delta t)\|_2 \leq 2C_q\,|\delta t|^{q+1}$.*

Proof. Because S_q is unitary, $\|R_q - \mathbb{I}\|_2 = \left\|US_q^\dagger - \mathbb{I}\right\|_2 = \|U - S_q\|_2 \leq C_q\,|\delta t|^{q+1}$. Diagonalize $R_q = W\,\mathrm{diag}(e^{-i\theta_j})W^\dagger$ with $\theta_j \in [-\pi, \pi]$ and define $K_q = W\,\mathrm{diag}(\theta_j)W^\dagger$. For any eigenphase, $\left|e^{-i\theta_j} - 1\right| = 2\,|\sin(\theta_j/2)|$. If $\|R_q - \mathbb{I}\|_2 \leq 1$, then $|\theta_j| \leq 2\arcsin(\|R_q - \mathbb{I}\|_2/2) \leq 2\,\|R_q - \mathbb{I}\|_2$. Taking the maximum over j proves the bound. □

Theorem 4 (Multi-step stability for approximate residuals). *Let $\widehat{R}_q$ be any approximate residual and define $\widehat{G}_q = \widehat{R}_q S_q$. If $\eta = \left\|\widehat{R}_q - R_q\right\|_2$, then*

$$\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \leq \eta\sum_{j=0}^{r-1}(1+\eta)^j \leq r\eta e^{(r-1)\eta}. \tag{6}$$

If $\widehat{R}_q$ is unitary, then the sharper bound $\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \leq r\eta$ holds.

Proof. Since S_q is unitary, $\left\|\widehat{G}_q - G_q\right\|_2 = \left\|(\widehat{R}_q - R_q)S_q\right\|_2 = \eta$. Also $G_q = U(\delta t)$ and $\|G_q\|_2 = 1$. The telescoping identity $\widehat{G}_q^{\,r} - G_q^r = \sum_{j=0}^{r-1}\widehat{G}_q^{\,r-1-j}(\widehat{G}_q - G_q)G_q^j$ combined with submultiplicativity gives Eq. (6), because $\left\|\widehat{G}_q\right\|_2 \leq \|G_q\|_2 + \eta = 1 + \eta$. If $\widehat{R}_q$ is unitary, then $\widehat{G}_q$ is unitary and every factor in the telescoping sum has norm one except $\widehat{G}_q - G_q$, giving the sharper bound. □

Theorem 5 (Tightness of the multi-step stability bound). *Let $\delta t > 0$, let $H = H^\dagger$, and let $S_q(\delta t)$ be an order-q step with exact residual R_q and corrected step $G_q = R_q S_q = U(\delta t)$. For every $\eta \in (0, \pi]$ there exists a* unitary *approximate residual $\widehat{R}_q$ with $\left\|\widehat{R}_q - R_q\right\|_2 \leq \eta$ such that, with $\widehat{G}_q = \widehat{R}_q S_q$,*

$$\left\|\widehat{G}_q^{\,r} - U(r\delta t)\right\|_2 \geq \tfrac{2}{\pi}\,r\eta \qquad \text{for every integer } r \text{ with } r\eta \leq \pi. \tag{7}$$

Hence the linear-in-r estimate of Theorem 4 is order-optimal: no bound of the form $o(r\eta)$ can hold uniformly in r.

Proof. Let $P = P^\dagger$ be a spectral projector of H onto a single eigenspace, so $\|P\|_2 = 1$ and $[P, U(s)] = 0$ for all real s. Set $\widehat{R}_q = e^{-i\eta P} R_q$, which is unitary. Since R_q is unitary, $\left\|\widehat{R}_q - R_q\right\|_2 = \left\|e^{-i\eta P} - \mathbb{I}\right\|_2 = \max_{p\in\{0,1\}} \left|e^{-i\eta p} - 1\right| = 2\sin(\eta/2) \le \eta$. Because $R_q S_q = U(\delta t)$ and $[P, U(\delta t)] = 0$,

$$\widehat{G}_q^{\,r} = \left(e^{-i\eta P} U(\delta t)\right)^r = e^{-ir\eta P} U(\delta t)^r = e^{-ir\eta P} U(r\delta t).$$

Therefore $\left\|\widehat{G}_q^{\,r} - U(r\delta t)\right\|_2 = \left\|e^{-ir\eta P} - \mathbb{I}\right\|_2 = 2\sin(r\eta/2)$. For $r\eta \le \pi$, Jordan's inequality $\sin x \ge \frac{2}{\pi} x$ on $[0, \pi/2]$ gives $2\sin(r\eta/2) \ge \frac{2}{\pi} r\eta$. □

Theorem 6 (Duhamel bound for generator approximation)**.** *Let K and $\widehat{K}$ be Hermitian. Then $\left\|e^{-iK} - e^{-i\widehat{K}}\right\|_2 \le \left\|K - \widehat{K}\right\|_2$. Consequently, if $R_q = e^{-iK_q}$ and $\widehat{R}_q = e^{-i\widehat{K}_q}$, then $\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \le r \left\|K_q - \widehat{K}_q\right\|_2$.*

Proof. Define $F(s) = e^{-i[(1-s)K + s\widehat{K}]}$. The derivative is represented by the standard Duhamel formula $\frac{dF}{ds} = -i \int_0^1 e^{-i(1-a)H_s} (\widehat{K} - K) e^{-iaH_s}\, da$, where $H_s = (1-s)K + s\widehat{K}$. Since H_s is Hermitian, the exponentials in the integral are unitary, so $\|dF/ds\|_2 \le \left\|\widehat{K} - K\right\|_2$. Integrating from $s = 0$ to $s = 1$ proves the first bound. The global bound follows from Theorem 4, since both exponentials are unitary. □

Definition 1 (Lie-algebraic spectral truncation)**.** *The Pauli-weight subspaces $\mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \cdots \subseteq \mathcal{B}_n = \mathsf{Herm}(d)$ form a filtration of the Lie algebra of Hermitian operators by interaction range. We call the orthogonal projection Π_w onto $\mathcal{B}_w$ (Eq. (5)) the* spectral truncation *at level w, and $\widehat{G}_{q,w} = e^{-i\Pi_w K_q} S_q$ the level-w truncated residual compilation. The level w is a single tunable knob interpolating from the uncorrected step ($w = 0$) to the exact correction ($w = n$); Theorem 7 and Corollary 1 show every level carries an* a priori *simulation certificate.*

Remark 2. *This is the Hamiltonian-simulation counterpart of spectral-truncation constructions in operator-algebraic machine learning, where truncating a noncommutative algebra yields a tunable family interpolating between local and global models with provable approximation control [73, 74]. Here the algebra is the Lie algebra of generators, the truncation knob is interaction range, and the approximation control is the multi-step simulation certificate rather than a kernel-approximation bound. Two consequences distinguish the simulation setting: the truncation converges* geometrically *in the level (Theorem 10), and the certificate is a bound on a downstream dynamical quantity, so the level is selectable from the target accuracy alone (Corollary 2).*

Theorem 7 (Frobenius-optimal Pauli compression)**.** *Let $K = K^\dagger$ on n qubits and let $\mathcal{B} \subseteq \mathrm{span}_{\mathbb{R}}(\mathcal{P}_n)$ be any real Pauli subspace. The orthogonal projection $\Pi_{\mathcal{B}} K = \sum_{P\in\mathcal{B}} \frac{\mathrm{Tr}(PK)}{2^n} P$ uniquely minimizes $\|K - L\|_F$ over $L \in \mathcal{B}$. The corresponding unitary residual approximation satisfies $\left\|e^{-iK} - e^{-i\Pi_{\mathcal{B}} K}\right\|_2 \le \|K - \Pi_{\mathcal{B}} K\|_2 \le \|K - \Pi_{\mathcal{B}} K\|_F$.*

Proof. The normalized Pauli operators $P/\sqrt{2^n}$ form an orthonormal basis under the Hilbert–Schmidt inner product. Orthogonal projection onto a closed finite-dimensional subspace is the unique Frobenius-norm minimizer. The spectral norm is bounded by the Frobenius norm, and Theorem 6 supplies the unitary perturbation bound. □

Corollary 1 (Projected residual simulation certificate)**.** *For $K_{q,w} = \Pi_w K_q$ and $\widehat{G}_{q,w} = e^{-iK_{q,w}} S_q$, $\left\| \widehat{G}^{\,r}_{q,w} - U(t) \right\|_2 \le r \left\| K_q - K_{q,w} \right\|_2$. Thus a target global accuracy ϵ is guaranteed whenever $\left\| K_q - K_{q,w} \right\|_2 \le \epsilon / r$.*

Proof. Both K_q and $K_{q,w} = \Pi_w K_q$ are Hermitian, the latter because Π_w is a real-linear projection onto a span of Hermitian Pauli strings (Theorem 7); hence $R_q = e^{-iK_q}$ and $\widehat{R}_{q,w} = e^{-iK_{q,w}}$ are both unitary. The Duhamel bound (Theorem 6) gives the per-step estimate $\left\| \widehat{R}_{q,w} - R_q \right\|_2 = \left\| e^{-iK_{q,w}} - e^{-iK_q} \right\|_2 \le \left\| K_q - K_{q,w} \right\|_2 =: \eta$. Since $\widehat{R}_{q,w}$ is unitary, the sharp branch of Theorem 4 applies to $\widehat{G}_{q,w} = \widehat{R}_{q,w} S_q$ and yields $\left\| \widehat{G}^{\,r}_{q,w} - U(t) \right\|_2 \le r\eta = r \left\| K_q - K_{q,w} \right\|_2$. Imposing $\eta \le \epsilon / r$ makes the right-hand side at most ϵ, which is the stated sufficient condition. $\square$

Corollary 2 (Certificate-driven bandwidth selection)**.** *Fix a target global accuracy $\epsilon > 0$, step count r, and step δt, and let $\tau_w := \left\| K_q - \Pi_w K_q \right\|_2$ be the truncation tail at level w. The quantities $\tau_0 \ge \tau_1 \ge \cdots \ge \tau_n = 0$ are nonincreasing and computable from K_q alone (no time evolution). The bandwidth*

$$w^\star(\epsilon) = \min\{\, w : \; r\,\tau_w \le \epsilon \,\} \tag{8}$$

is well defined and the level-$w^\star$ truncated compilation meets the target, $\left\| \widehat{G}^{\,r}_{q,w^\star} - U(t) \right\|_2 \le \epsilon$. If, moreover, $\tau_w \le C\rho^w$ with $\rho < 1$ (the geometric regime of Theorem 10), then $w^\star(\epsilon) \le \log(rC/\epsilon)/\log(1/\rho)$ grows only logarithmically in $1/\epsilon$.

Proof. Monotonicity of τ_w holds because $\mathcal{B}_w \subseteq \mathcal{B}_{w+1}$, so the orthogonal projection onto the larger subspace cannot increase the residual norm; $\tau_n = 0$ because $\mathcal{B}_n = \mathsf{Herm}(d)$ and $\Pi_n K_q = K_q$. Hence the set $\{w : r\tau_w \le \epsilon\}$ contains $w = n$ and has a least element $w^\star$. The accuracy guarantee is Corollary 1 with $w = w^\star$, since $r\tau_{w^\star} \le \epsilon$. For the logarithmic bound, $r\tau_w \le rC\rho^w \le \epsilon$ as soon as $w \ge \log(rC/\epsilon)/\log(1/\rho)$, and $w^\star$ is the least such integer. $\square$

Theorem 8 (Leading Strang residual support for the open-boundary TFIM)**.** *Let A and B be the TFIM splitting in Eq. (3). For the Strang step $S_2(\delta t)$, the degree-three homogeneous term in the residual generator $K_2(\delta t)$ is a real linear combination of Pauli strings with weight at most three. Equivalently,*

$$K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5), \qquad K_2^{(3)} \in \mathcal{B}_3. \tag{9}$$

Proof. The symmetric BCH expansion of Strang splitting has no degree-two term; the first nonzero defect is a linear combination of double commutators $[A,[A,B]]$ and $[B,[A,B]]$ [42, 43]. The residual generator is the negative of this leading logarithmic defect up to the same order, so it is enough to bound the Pauli weight of these double commutators.

Write $A = \sum_e A_e$ with $A_e = JZ_jZ_{j+1}$ for edges $e = (j, j+1)$ and $B = \sum_v B_v$ with $B_v = hX_v$. A commutator $[A_e, B_v]$ vanishes unless v is incident to e. If it is nonzero, its support is contained in the two sites of e and has Pauli weight two. In $[B,[A,B]]$, the second commutator adds at most one on-site term $B_{v'}$ whose support must overlap the existing support; therefore the support remains within the same edge and the Pauli weight is at most two. In $[A,[A,B]]$, the second edge term $A_{e'}$ must overlap the support of $[A_e, B_v]$. The union of two overlapping nearest-neighbor edges in a one-dimensional open chain contains at most three sites. Therefore every nonzero Pauli string in the double commutators has weight at most three. This proves $K_2^{(3)} \in \mathcal{B}_3$. $\square$

Theorem 9 (Order generality of residual locality)**.** *Let $A = J\sum_e Z_jZ_{j+1}$ and $B = h\sum_v X_v$ be the open-boundary TFIM splitting, and let S_q be a symmetric product formula of order q. The leading term of its residual generator is geometrically local of bounded Pauli weight:*

$$K_q(\delta t) = \delta t^{q+1} K_q^{(q+1)} + O(\delta t^{q+3}), \qquad K_q^{(q+1)} \in \mathcal{B}_{q+2}. \tag{10}$$

Proof. The leading defect of a symmetric order-q product formula is $O(\delta t^{q+1})$, and by the symmetric Baker–Campbell–Hausdorff / Magnus structure its operator coefficient is a real linear combination of (q+1)-fold nested commutators of A and B [42, 43]; the residual generator cancels this leading logarithmic defect, so $K_q^{(q+1)}$ is itself such a linear combination. We bound the Pauli weight of any nonzero m-fold nested commutator of the TFIM local terms by induction on the number m of local factors. A single factor $A_e = JZ_jZ_{j+1}$ or $B_v = hX_v$ has support of at most two sites, so weight $\le 2 = m+1$ at $m = 1$. Assume an (m−1)-fold nested commutator C has support on at most m sites. A further commutator $[L, C]$ with a local factor L of range at most two is nonzero only if $\mathrm{supp}(L) \cap \mathrm{supp}(C) \neq \emptyset$, and $\mathrm{supp}([L, C]) \subseteq \mathrm{supp}(L) \cup \mathrm{supp}(C)$, a set of at most $|\mathrm{supp}(C)| + 2 - 1 \le m + 1$ sites. Hence every nonzero m-fold nested commutator has Pauli weight at most $m + 1$, and the leading (q+1)-fold term has weight at most $q + 2$, i.e. $K_q^{(q+1)} \in \mathcal{B}_{q+2}$. □

Remark 3. *Theorem 9 is the worst-case algebraic bound; in the symmetric formula many maximal-weight strings cancel, so the* effective *compression threshold is lower. Table 2 measures the dominant collapse at weight* $3, 4, 5$ *for* $q = 2, 4, 6$*, i.e.* $\approx \lceil q/2\rceil + 2$*, comfortably inside the proven* $q + 2$ *envelope and tight at* $q = 2$*, where Theorem 8 gives the sharper bound three.*

Theorem 10 (Geometric convergence rate of the spectral truncation)**.** *Let* A, B *be the open-boundary TFIM splitting and* S_q *the symmetric order-q step. There exist constants* $C_q > 0$*, a ratio* $\rho \in (0, 1)$ *with* $\rho = O(\delta t)$*, and* $\delta t_0 > 0$ *such that, for* $\delta t \le \delta t_0$ *and every integer* w *with* $q + 1 \le w \le n$*, the level-w spectral-truncation tail* $\tau_w = \|K_q - \Pi_w K_q\|_2$ *obeys the* geometric envelope

$$\tau_w \le C_q\, \rho^{w}. \tag{11}$$

Consequently, by Corollary 1, the certified global error $r\tau_w$ *contracts by a factor* $\rho = O(\delta t)$ *per added level,* $\left\|\widehat{G}^{r}_{q,w} - U(t)\right\|_2 \le r\, C_q\, \rho^{w}$*.*

Proof. The residual factor $R_q(\delta t) = U(\delta t) S_q(\delta t)^\dagger$ is real-analytic in δt and equals $\mathbb{I}$ at $\delta t = 0$, so for $|\delta t|$ below some radius $\delta t_0 > 0$ it remains in the domain on which the principal logarithm is analytic; hence $K_q(\delta t) = i \log R_q(\delta t)$ is real-analytic near 0 and its homogeneous expansion $K_q(\delta t) = \sum_{d\ge q+1} \delta t^{\,d} K_q^{(d)}$ converges absolutely for $|\delta t| \le \delta t_0$, with $M := \sum_{d\ge q+1} \delta t_0^{d} \left\|K_q^{(d)}\right\|_2 < \infty$. (Time-reversal symmetry of S_q makes only the degrees $q + 1, q + 3, \ldots$ appear [43], which we do not need beyond convergence.) By the nested-commutator induction of Theorem 9, $K_q^{(d)} \in \mathcal{B}_{d+1}$, so any Pauli string of weight $\ge w + 1$ first occurs at degree $d \ge w$; the part of K_q outside $\mathcal{B}_w$ is therefore a sum of homogeneous terms of degree $d \ge w$, and for $\delta t \le \delta t_0$,

$$\tau_w = \Big\|\sum_{d\ge w} \delta t^{\,d}\, \Pi_{>w} K_q^{(d)}\Big\|_2 \le \sum_{d\ge w} \delta t^{\,d} \left\|K_q^{(d)}\right\|_2 = \left(\tfrac{\delta t}{\delta t_0}\right)^{w} \sum_{d\ge w} \left(\tfrac{\delta t}{\delta t_0}\right)^{d-w} \delta t_0^{d} \left\|K_q^{(d)}\right\|_2 \le M \left(\tfrac{\delta t}{\delta t_0}\right)^{w},$$

using $(\delta t/\delta t_0)^{d-w} \le 1$ for $d \ge w$. This is Eq. (11) with $C_q = M$ and $\rho = \delta t/\delta t_0 = O(\delta t)$. The global estimate follows from Corollary 1. □

Remark 4. *Equation (11) is a conservative provable envelope (per-level contraction* $\rho = O(\delta t)$*, i.e. a factor* $\gtrsim \delta t^{-1} = 10$*). The measured per-level factors are far larger—*$449\times$ *for* $q = 2$ *(least-squares fit over* $w = 3, 4, 5$*) and* $576\times$ *for* $q = 4$ *(the single resolvable pair* $w = 4, 5$ *before the round-off floor) at* $\delta t = 0.1$ *(Table 3)—because the symmetric cancellations that lower the effective locality threshold to* $\approx \lceil q/2\rceil + 2$ *(Remark after Theorem 9) also raise the degree at which each new weight level first appears, so empirically each retained level absorbs content roughly two powers of* δt *smaller, i.e. an effective ratio* $\approx \delta t^2$ *further amplified by prefactors. The provable envelope already gives a geometric* rate, *not merely convergence in a limit, which is what makes the level a usable accuracy dial (Corollary 2).*

Theorem 11 (Faithful gate-level compilation of the correction)**.** *Let the compressed generator be partitioned into m layers $\widehat{K}_q = \sum_{i=1}^{m} L_i$, where each L_i is a real linear combination of* mutually commuting *Pauli strings, and let $\widehat{R}_q^{\mathrm{c}} = \prod_{i=1}^{m} e^{-iL_i}$ be the first-order compiled correction. Then, with $\Lambda = \sum_{i=1}^{m} \|L_i\|_2$,*

$$\left\| \widehat{R}_q^{\mathrm{c}} - e^{-i\widehat{K}_q} \right\|_2 \le \tfrac{1}{2} \sum_{i<j} \|[L_i, L_j]\|_2 \le \tfrac{1}{2}\Lambda^2. \tag{12}$$

Consequently $\widehat{G}_q^{\mathrm{c}} = \widehat{R}_q^{\mathrm{c}} S_q$ obeys, for $t = r\delta t$,

$$\left\| (\widehat{G}_q^{\mathrm{c}})^r - U(t) \right\|_2 \le r \left\| K_q - \widehat{K}_q \right\|_2 + \tfrac{1}{2}\, r\, \Lambda^2. \tag{13}$$

Since every coefficient of $\widehat{K}_q$ is $O(\delta t^{q+1})$ (Theorem 3), $\Lambda = O(\delta t^{q+1})$ and the compilation term is $O(r\,\delta t^{2(q+1)})$.

Proof. The first inequality is the standard first-order Lie–Trotter estimate for the difference between the ordered product $\prod_i e^{-iL_i}$ and $e^{-i\sum_i L_i}$, bounded to second order by $\frac{1}{2}\sum_{i<j} \|[L_i, L_j]\|_2$; using $\|[L_i, L_j]\|_2 \le 2\|L_i\|_2 \|L_j\|_2$ and $\sum_{i<j} 2\|L_i\|_2 \|L_j\|_2 \le (\sum_i \|L_i\|_2)^2 = \Lambda^2$ gives the $\frac{1}{2}\Lambda^2$ bound. The global estimate follows by the triangle inequality $\left\| \widehat{R}_q^{\mathrm{c}} - R_q \right\|_2 \le \left\| \widehat{R}_q^{\mathrm{c}} - e^{-i\widehat{K}_q} \right\|_2 + \left\| e^{-i\widehat{K}_q} - e^{-iK_q} \right\|_2$, the Duhamel bound (Theorem 6) on the second term, and the unitary multi-step bound (Theorem 4) applied with both $\widehat{R}_q^{\mathrm{c}}$ and R_q unitary. Finally each $\|L_i\|_2 \le \sum_{P \in L_i} |c_P|$ with c_P the Pauli coefficients of $\widehat{K}_q$, so $\Lambda \le \sum_P |c_P| = O(\delta t^{q+1})$. □

Remark 5. *Theorem 11 is why the frontier advance is specific to $q \ge 4$. The corrected step's intrinsic error after removing the leading defect is $O(\delta t^{q+3})$, while the inner-compilation error is $O(\delta t^{2(q+1)})$; the latter is asymptotically smaller exactly when $2(q+1) > q+3$, i.e. $q > 1$, but the* prefactors *matter at finite δt: at $q = 2$, $\delta t = 0.1$ the compilation term ($\sim \Lambda^2$ with $\Lambda \sim \|K_2\| \sim 10^{-3}$) caps the compiled $w \le 4$ correction near 2×10^{-5} (Table 6), whereas at $q = 4$, $\Lambda \sim \|K_4\| \sim 10^{-6}$ and the compilation term is $\sim 10^{-12}$, far below the corrected accuracy. The theory thus predicts, and the data confirm, that faithful compilation requires a sufficiently small base-step residual—i.e. a sufficiently high base order.*

Theorem 12 (Symmetric faithful compilation)**.** *With the layer partition $\widehat{K}_q = \sum_{i=1}^{m} L_i$ of Theorem 11, let the* symmetric *compiled correction be the palindromic product*

$$\widehat{R}_q^{\mathrm{s}} = \Big(\prod_{i=1}^{m} e^{-iL_i/2} \Big) \Big(\prod_{i=m}^{1} e^{-iL_i/2} \Big). \tag{14}$$

Then, with $\Lambda = \sum_i \|L_i\|_2$ and a constant c depending only on m,

$$\left\| \widehat{R}_q^{\mathrm{s}} - e^{-i\widehat{K}_q} \right\|_2 \le c\,\Lambda^3, \tag{15}$$

and consequently $\widehat{G}_q^{\mathrm{s}} = \widehat{R}_q^{\mathrm{s}} S_q$ obeys $\left\| (\widehat{G}_q^{\mathrm{s}})^r - U(t) \right\|_2 \le r \left\| K_q - \widehat{K}_q \right\|_2 + c\, r\, \Lambda^3$. Since $\Lambda = O(\delta t^{q+1})$, the inner-compilation term is $O(r\,\delta t^{3(q+1)})$—one order of $(q+1)$ smaller than the first-order bound $O(r\,\delta t^{2(q+1)})$ of Theorem 11—at the cost of applying each layer twice, with the two central half-layers merging, so the correction's two-qubit-gate count rises from $\sum_i \mathrm{cost}(L_i)$ to $2\sum_i \mathrm{cost}(L_i) - \mathrm{cost}(L_m)$.

Proof. $\widehat{R}_q^{\mathrm{s}}$ is the symmetric (Strang) composition of the one-parameter unitaries e^{-isL_i}, a time-symmetric second-order integrator for the flow of $-i\sum_i L_i = -i\widehat{K}_q$ over unit time. By the standard backward-error / symmetric-BCH analysis of such compositions [43], the leading commutator defect of the first-order ordered product cancels by symmetry and the remaining error is third order in the generators, $\left\|\widehat{R}_q^{\mathrm{s}} - e^{-i\widehat{K}_q}\right\|_2 \le c\sum_{i\le j\le k}\|[L_i,[L_j,L_k]]\|_2 \le c'\Lambda^3$, absorbing c' into c. The global estimate follows as in Theorem 11: triangle inequality through $e^{-i\widehat{K}_q}$, the Duhamel bound (Theorem 6) on $\left\|e^{-i\widehat{K}_q} - e^{-iK_q}\right\|_2$, and the unitary multi-step bound (Theorem 4). The gate count follows because each layer L_i appears as two half-angle exponentials of identical CNOT cost to the full-angle one, except the central pair $e^{-iL_m/2}e^{-iL_m/2} = e^{-iL_m}$, which merges into a single layer. □

Remark 6. *Theorem 12 removes the structural floor that Theorem 11 leaves at low base order. At $n = 6$, $q = 2$, $w \le 4$ the first-order compile stalls at 2.18×10^{-5} (Table 6) against the dense-oracle 3.65×10^{-7} (Table 4); the symmetric compile reaches 3.76×10^{-7}, recovering the oracle, at roughly twice the correction's CNOT count (Table 4). This is the simulation counterpart of the smoothing that restores positive-definiteness in operator-algebraic spectral truncation [73]: a symmetry-motivated modification of the naive realization that repairs the truncated object's structural defect. At $q = 4$ the first-order compile is already faithful and cheaper, so the headline frontier advance (Sec. 1.7) does not need the symmetric compile; its role is to make the compress-then-compile pipeline faithful at* every *base order.*

Theorem 13 (Size-transfer error bound for learned local residual generators)**.** *Consider the disordered open-boundary TFIM on n sites with couplings in a fixed compact set, and let $K_2^{(3)}(n) = \sum_{j=1}^n h_j$ be the degree-three Strang residual generator, where by Theorem 8 each h_j is a real linear combination of weight-≤ 3 Pauli strings supported in the three-site window $\{j-1, j, j+1\}$ (truncated at the boundaries). Suppose a translation-invariant per-site predictor outputs $\widehat{h}_j$ from the couplings in that window with $\left\|\widehat{h}_j - h_j\right\|_2 \le \eta$ for every j, where η depends only on the predictor and the coupling range, not on n. Then $\widehat{K}_2^{(3)}(n) = \sum_j \widehat{h}_j$ satisfies $\left\|\widehat{K}_2^{(3)}(n) - K_2^{(3)}(n)\right\|_2 \le n\,\eta$, and the corrected step $\widehat{G}(\delta t) = \exp(-i\delta t^3\widehat{K}_2^{(3)})\,S_2(\delta t)$ obeys, for $t = r\delta t$, $\left\|\widehat{G}(\delta t)^r - U(t)\right\|_2 \le r\,\delta t^3 n\,\eta + r\,O(\delta t^5)$.*

Proof. The first bound is the triangle inequality over the n site-anchored terms,

$$\left\|\textstyle\sum_j(\widehat{h}_j - h_j)\right\|_2 \le \sum_j \left\|\widehat{h}_j - h_j\right\|_2 \le n\eta;$$

translation invariance makes the per-site bound η independent of n. For the second, set $\widehat{K} = \delta t^3\widehat{K}_2^{(3)}$. The BCH expansion in the Extended Data (detailed TFIM support analysis) gives $K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5)$, so $\left\|\widehat{K} - K_2(\delta t)\right\|_2 \le \delta t^3 n\eta + O(\delta t^5)$. The Duhamel bound (Theorem 6) gives $\left\|e^{-i\widehat{K}} - R_2\right\|_2 \le \left\|\widehat{K} - K_2\right\|_2$, and the multi-step stability bound of Theorem 4 multiplies by r, giving the stated estimate. □

Theorem 14 (From per-site coefficient error to a global certificate)**.** *Adopt the setting of Theorem 13 and expand each per-site generator in the fixed family of T weight-≤ 3 Pauli templates anchored at site j, $h_j = \sum_{\alpha=1}^T c_{j,\alpha}P_{j,\alpha}$ and $\widehat{h}_j = \sum_{\alpha=1}^T \widehat{c}_{j,\alpha}P_{j,\alpha}$, where each $P_{j,\alpha}$ is a Pauli string, so $\|P_{j,\alpha}\|_2 = 1$. If the per-site coefficient error obeys $\|c_j - \widehat{c}_j\|_2 \le \delta_c$ for every j, then the per-site operator error of Theorem 13 satisfies $\eta \le \sqrt{T}\,\delta_c$, and the learned corrected step $\widehat{G}(\delta t) = \exp(-i\delta t^3\widehat{K}_2^{(3)})\,S_2(\delta t)$ obeys, for $t = r\delta t$,*

$$\left\|\widehat{G}(\delta t)^r - U(t)\right\|_2 \le r\,\delta t^3\,n\sqrt{T}\,\delta_c + r\,O(\delta t^5). \tag{16}$$

Proof. By the triangle inequality and $\|P_{j,\alpha}\|_2 = 1$,

$$\left\|\widehat{h}_j - h_j\right\|_2 = \left\|\sum_{\alpha=1}^{T}(\widehat{c}_{j,\alpha} - c_{j,\alpha})P_{j,\alpha}\right\|_2 \leq \sum_{\alpha=1}^{T}|\widehat{c}_{j,\alpha} - c_{j,\alpha}| = \|c_j - \widehat{c}_j\|_1 \leq \sqrt{T}\,\|c_j - \widehat{c}_j\|_2 \leq \sqrt{T}\,\delta_c,$$

where the last line uses the Cauchy–Schwarz inequality $\|x\|_1 \leq \sqrt{T}\,\|x\|_2$ on $\mathbb{R}^T$. Taking the maximum over j gives $\eta \leq \sqrt{T}\,\delta_c$; substituting this into the size-transfer bound of Theorem 13 yields Eq. (16). □

Remark 7. *With the $T = 48$ contiguous weight-≤ 3 templates per site used in Sec. 1.9 and the measured held-out coefficient accuracy (relative ℓ_2 error 7.2×10^{-3}, $R^2 = 0.9999$), Theorem 14 converts the learned-coefficient accuracy directly into the* a priori *global error bound whose linear-in-r growth Fig. 5c verifies empirically.*

Remark 8. *Theorem 8 does not say the full residual generator lies in $\mathcal{B}_3$. Higher-order BCH terms can produce larger support. It predicts that $w \leq 3$ projection should remove the dominant Strang defect at small step size, while $w \leq 4$ and higher should capture additional subleading structure. This is consistent with the numerical results reported above.*

Theorem 15 (A posteriori step certificate)**.** *Let $\widehat{G}$ be any unitary approximate one-step propagator and let $\Delta = \left\|\widehat{G} - U(\delta t)\right\|_2$. Then $\left\|\widehat{G}^r - U(t)\right\|_2 \leq r\Delta$.*

Proof. Apply the telescoping identity to $\widehat{G}^r - U(\delta t)^r$. Both $\widehat{G}$ and $U(\delta t)$ are unitary, so each of the r terms is bounded by Δ. □

3.3 Numerical methodology

All numerical results are generated by deterministic dense-matrix scripts that build the Pauli matrices, construct A, B, and H, evaluate matrix exponentials using dense linear algebra, compute spectral norms from singular values, and write the resulting CSV, JSON, PDF, and table files. There are no hand-coded performance numbers in the plotting pipeline. The deterministic studies use fixed grids with no stochastic sampling. The learned-residual experiment (Sec. 1.9) does draw disordered couplings and train a network, but it is made reproducible by fixed random seeds for the sampler, the data split, and the network initialization; its training labels and all reported error metrics are computed by the same dense-matrix code, so rerunning the script reproduces every reported value. The benchmark design deliberately includes checks that would expose fabrication or implementation errors: unitarity checks for S_q, R_q, and corrected steps; same-order comparisons between S_q and R_qS_q; exact dense recomputation of $U(t)$ rather than relying on repeated one-step products as the reference; saved raw data for every plotted point; and finite-precision reporting, including cases where round-off makes the high-order baseline comparable to the oracle residual.

The Hilbert-space dimension is $d = 2^n$. For the fixed-time study we use $J = h = 1$, $t = 1$, and $r = 10$ steps. The exact total propagator $U(t)$ is computed as $\exp(-iHt)$ independently of the product-formula steps. For the projected residual study we compute the residual generator $K_q = i\log(R_q)$ using the dense matrix logarithm, symmetrize K_q as $(K_q + K_q^\dagger)/2$ to remove round-off skew-Hermitian contamination, and project onto Pauli strings of weight at most w using Eq. (5). The projected corrected propagator is $[e^{-i\Pi_w K_q}S_q]^r$. The learned-residual study additionally trains small multilayer perceptrons (on global-oracle and on oracle-free local-patch labels) with a standard automatic-differentiation library on disordered chains; all couplings, step sizes, and network weights are seeded from a single fixed seed, and every learned generator is scored only through the dense spectral-norm metrics above. Each network is trained for 4000 full-batch Adam steps on 220 disordered realizations at each of $n = 4$ and $n = 5$; reported transfer errors are averaged over independent held-out realizations at each size (40 realizations for $n = 4$–7, 24 for $n = 8$, 16 for $n = 9$, 12 for $n = 10$), and the shaded bands in Fig. 5 report one standard deviation across these realizations. All plotted errors are spectral-norm errors. Where a plotted oracle residual appears around 10^{-14} to 10^{-13}, that value is the observed double-precision floor, not a mathematical error in exact arithmetic.

Two-qubit-gate cost model. The error–cost frontier (Sec. 1.7, Fig. 4, Table 6) uses the standard textbook compilation of Pauli exponentials into Clifford-plus-rotation gates, accounting only two-qubit (CNOT) gates as is conventional in fault-tolerant and near-term resource estimates. For the TFIM, $e^{-iB\tau}$ is a layer of single-qubit X rotations (no two-qubit gates), and $e^{-iA\tau}$ is a layer of $n-1$ nearest-neighbour ZZ rotations, each costing two CNOTs, so an A-layer costs $2(n-1)$ CNOTs; a Suzuki step S_q with $\#A_q$ elementary A-exponentials costs $2(n-1)\,\#A_q$ CNOTs, with $\#A_q = 1, 5, 25, 125$ for $q = 2, 4, 6, 8$. A weight-w Pauli rotation $e^{-i\theta P}$ compiles by a CNOT ladder into $2(w-1)$ CNOTs. The compiled correction is partitioned into mutually-commuting layers by greedy colouring of the anticommutation graph (Theorem 11), and *every reported corrected-step error is the error of this compiled correction*, not of the dense oracle; its CNOT cost is the sum of $2(w-1)$ over the retained Pauli terms. For the symmetric inner compilation (Theorem 12, Table 4) each commuting layer is applied twice at half angle with the two central half-layers merged, so the correction cost rises to $2\sum_i \mathrm{cost}(L_i) - \mathrm{cost}(L_m)$ over the m layers; we report both costs. Single-qubit gates are not counted; the classical cost of obtaining the generator is offline and, in the learned mode (Sec. 1.9), amortized and free of any dense 2^n propagator.

Data availability

All data supporting the numerical findings are generated by deterministic scripts from the Hamiltonian definition in Eq. (3). The raw CSV and JSON files, figure-generation scripts, and table-generation scripts are included with the manuscript package. No experimental result is fabricated or manually inserted without a corresponding generated data record.

Code availability

The accompanying code is a pip-installable package (`lieideal-hs`) that constructs Pauli operators, Trotter–Suzuki steps, exact propagators, residual generators, spectral truncations, the symmetric and first-order compilations, the certificate-driven bandwidth selection, the learned per-site model, and every table and figure from first principles. Installing it and running a single console entry point (`lieideal-reproduce`) deterministically regenerates every dataset, table, and figure of this manuscript; a companion entry point (`lieideal-verify`) reruns the pipeline and asserts byte-identical outputs. The spectral-truncation convergence rate, the certificate, and the symmetric-versus-first-order compilation are produced by the `spectral_truncation` module. The code is intended as a reproducibility baseline and as a starting point for GPU-accelerated implementations.

Author contributions

M.H. conceived the residual-generator framework, developed and proved the theoretical results, designed and implemented the dense-matrix and operator-learning experiments, produced all figures and tables, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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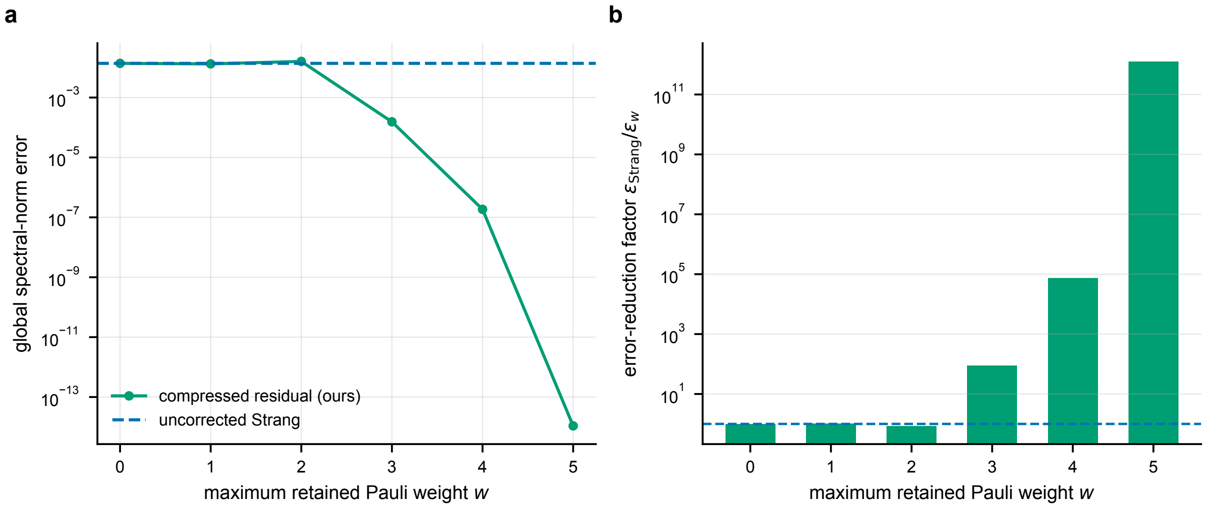


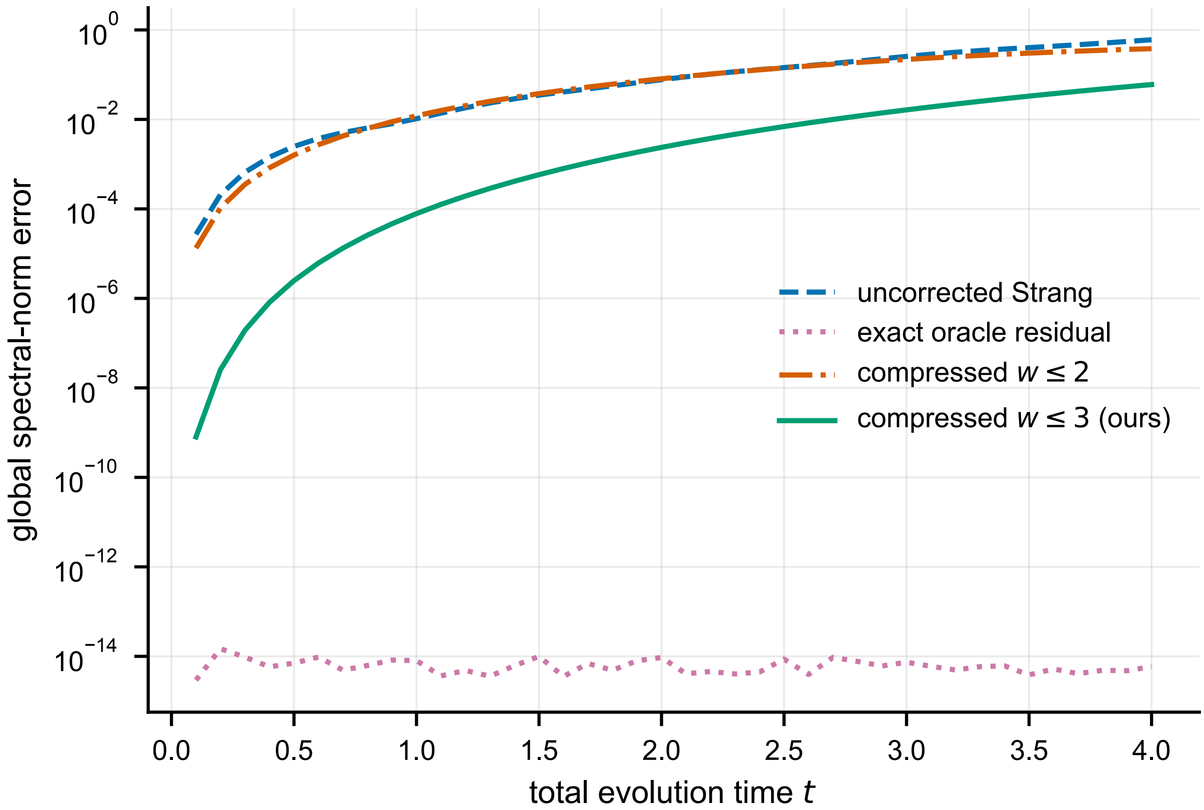
Extended Data Fig. 1 | Compressed-residual compilation for $n = 5$, $q = 2$ ($J = h = 1$, $t = 1$, $r = 10$). **a**, Spectral-norm error versus maximum retained Pauli weight w, against the uncorrected baseline. **b**, Improvement factor over the baseline by weight. The sharp improvement at $w = 3$ is predicted by the leading-support theorem for the Strang residual (Theorem 8). All values are exact, deterministic dense-matrix computations (no sampling, hence no error bars); the $w = 5$ row sits at the double-precision floor.

Table 9. Weight-truncated residual compilation for $n = 5$, $q = 2$, $J = h = 1$, $t = 1$, and $r = 10$. The column w keeps Pauli strings of weight at most w in the exact residual generator. Every entry is a single exact, deterministic dense-matrix computation (no statistical sampling, hence no uncertainty interval); the $w = 5$ row is at the double-precision floor.

w	$\|K_q - \Pi_w K_q\|_2$	projected error	improvement
0	3.656×10^{-3}	1.384×10^{-2}	$1.00\times$
1	2.859×10^{-3}	1.330×10^{-2}	$1.04\times$
2	1.949×10^{-3}	1.607×10^{-2}	$0.86\times$
3	1.573×10^{-5}	1.545×10^{-4}	$89.56\times$
4	1.850×10^{-8}	1.850×10^{-7}	$74829.05\times$
5	1.143×10^{-18}	1.089×10^{-14}	$1270805509048.55\times$

Extended Data

The per-size breakdown of every correction (Extended Data Table 12) makes the overlap of the oracle-trained and oracle-free learned curves in Fig. 5a quantitative: at every size the oracle-free network matches or slightly improves on the oracle-trained one and both sit about a factor of five below the analytic leading-order correction, while staying within a small constant factor of the exact weight-≤ 3 oracle. The single-step step-size sweep at $n = 6$ (Extended Data Table 13) tabulates the numbers behind Fig. 5b: the analytic δt^3 correction degrades from an $11\times$ improvement over the uncorrected step at $\delta t = 0.05$ to only $1.24\times$ at $\delta t = 0.30$, whereas the δt-conditioned learned correction stays close to the exact oracle, closing from about a factor of fourteen at $\delta t = 0.05$ to within roughly 20% at $\delta t = 0.30$ and remaining far below the analytic baseline throughout. The residual-generator norms in Extended Data Table 11 confirm the order scaling underlying Theorem 3: increasing the order from $q = 1$ to $q = 6$ shrinks $\|K_q(0.1)\|_2$ from 4.44×10^{-2} to 1.43×10^{-9}.

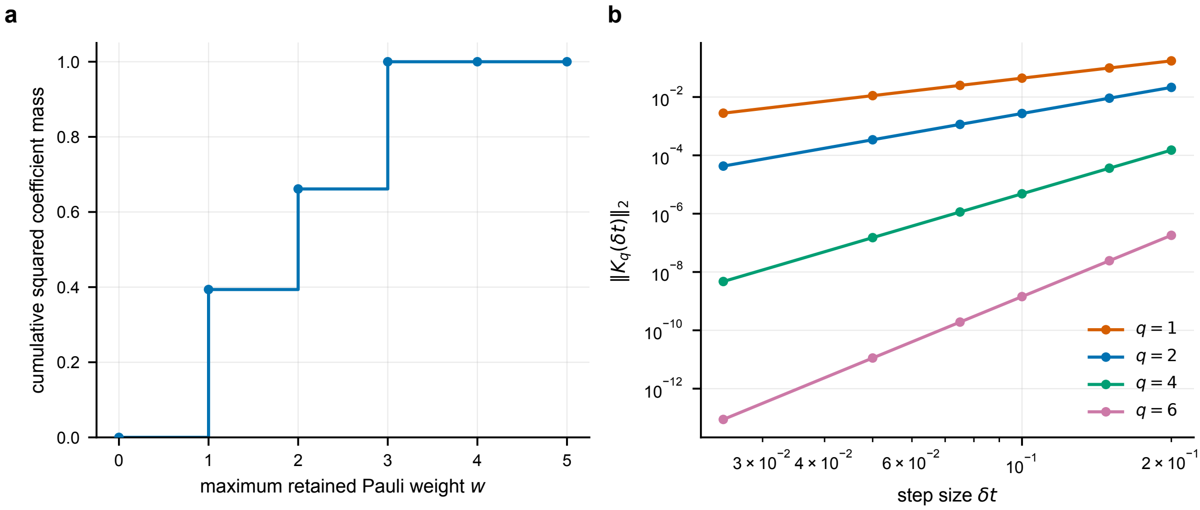


Extended Data Fig. 2 | Time-resolved correction hierarchy for $n = 4$, $q = 2$, $J = h = 1$, and $r = 10$, over 40 total times $t \in [0.1, 4.0]$. The $w \le 3$ projected residual uniformly improves the baseline in this sweep; $w \le 2$ is less reliable. Each curve is an exact, deterministic dense-matrix computation (no sampling, hence no error bars).

Supplementary theory: detailed BCH support analysis for the TFIM

The support theorem in the main text is intentionally stated without relying on exact BCH coefficients. This appendix gives the more explicit structure. For Strang splitting, $S_2(\delta t) = e^{-iB\delta t/2}e^{-iA\delta t}e^{-iB\delta t/2}$, the logarithm has the form $\log S_2(\delta t) = -i(A+B)\delta t + \delta t^3\Gamma_3 + O(\delta t^5)$, where Γ_3 is a linear combination of $[A,[A,B]]$ and $[B,[A,B]]$. The absence of an even power follows from time-reversal symmetry of the Strang formula. The residual is $R_2(\delta t) = e^{-i(A+B)\delta t}S_2(\delta t)^\dagger$, and the leading residual generator cancels the leading logarithmic defect, so $K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5)$, where $K_2^{(3)}$ is a real linear combination of $i[A,[A,B]]$ and $i[B,[A,B]]$ after accounting for the anti-Hermitian nature of commutators of Hermitian matrices.

For the TFIM, write $A_e = JZ_jZ_{j+1}$ and $B_v = hX_v$. The elementary commutator is $[A_e, B_v] = 0$ if $v \notin e$. If $v = j$, then $[Z_jZ_{j+1}, X_j] = 2iY_jZ_{j+1}$, and if $v = j+1$, then $[Z_jZ_{j+1}, X_{j+1}] = 2iZ_jY_{j+1}$. Thus $[A,B]$ is a sum of weight-two Pauli strings on nearest-neighbor edges. Next consider $[B,[A,B]]$: a term $[B_{v'},[A_e,B_v]]$ is nonzero only if v' lies in the support of $[A_e,B_v]$, which is the same edge e; the resulting Pauli string remains supported on that edge and has weight at most two, for example $[X_j, Y_jZ_{j+1}] = 2iZ_jZ_{j+1}$ and $[X_{j+1}, Y_jZ_{j+1}] = -2iY_jY_{j+1}$. Now consider $[A,[A,B]]$: a term $[A_{e'},[A_e,B_v]]$ is nonzero only if e' overlaps the support of $[A_e,B_v]$. In a one-dimensional nearest-neighbor chain, the union of two overlapping edges contains at most three sites, $(j-1, j, j+1)$ or $(j, j+1, j+2)$. Therefore every nonzero Pauli string in $[A,[A,B]]$ has weight at most three, proving that the leading Strang residual generator is contained in $\mathcal{B}_3$. The same proof strategy extends to other local Hamiltonians: if the interaction



Extended Data Fig. 3 | Structure of the residual generator. **a**, Cumulative squared Pauli coefficient mass for the $n = 5$, $q = 2$ residual generator at $\delta t = 0.1$; weight three captures the dominant leading structure, while higher weights capture subleading BCH terms. **b**, Residual-generator norm $\|K_q(\delta t)\|_2$ versus step size for $n = 4$ and $q = 1, 2, 4, 6$; the norm decreases rapidly with order, consistent with product-formula order conditions and Theorem 3. Both panels are exact, deterministic dense-matrix computations (no sampling, hence no error bars).

Table 10. Per-step resource proxy. The baseline factor count is the number of elementary $e^{-iA\tau}$ or $e^{-iB\tau}$ exponentials in S_q. The oracle residual construction uses one additional dense exponential for $U(\delta t)$ and one dense multiplication by $S_q^\dagger$; it is not a scalable gate-level implementation. All entries are exact integer counts, not measured quantities.

q	Suzuki factors	dense $U(\delta t)$ expm	dense residual multiply
1	2	1	1
2	3	1	1
4	15	1	1
6	75	1	1
8	375	1	1

hypergraph has local terms of diameter at most r_0, then an m-fold nested commutator can only grow along overlapping local supports, connecting residual-generator compression to locality bounds and Lieb–Robinson ideas [29, 30].

Supplementary theory: projection formulas and diagnostic quantities

For n qubits, the Pauli group modulo phases gives 4^n Hermitian strings satisfying $\mathrm{Tr}(PQ) = 2^n\delta_{P,Q}$. Every Hermitian matrix K has an expansion $K = \sum_{P\in\mathcal{P}_n} c_P P$ with $c_P = \mathrm{Tr}(PK)/2^n$, squared Frobenius norm $\|K\|_F^2 = 2^n \sum_{P\in\mathcal{P}_n} |c_P|^2$, and cumulative coefficient mass (plotted in Extended Data Fig. 3a) $M(w) = \sum_{\mathrm{wt}(P)\le w} |c_P|^2 / \sum_{P\in\mathcal{P}_n} |c_P|^2$. For the $n = 5$, $q = 2$, $\delta t = 0.1$ run, the generated squared coefficient masses by weight are $M_0' = 3.598 \times 10^{-33}$, $M_1' = 1.466 \times 10^{-6}$, $M_2' = 9.973 \times 10^{-7}$, $M_3' = 1.262 \times 10^{-6}$, $M_4' = 1.235 \times 10^{-10}$, and $M_5' = 3.413 \times 10^{-16}$. Although the weight-four mass is small in Frobenius terms, it can still matter for spectral-norm accuracy after repeated steps, which is why Extended Data Table 9 shows a large improvement from $w = 3$ to $w = 4$.

Table 11. Residual-generator spectral norm versus step size and product-formula order. $\|K_q(\delta t)\|_2$ for the open-boundary TFIM at $n = 4$, $J = h = 1$, computed by dense matrix logarithm of the exact residual factor. The norm falls steeply with order q and scales as δt^{q+1} at small δt, the quantitative content of Theorem 3 and Extended Data Fig. 3b. Each entry is a single exact, deterministic dense-matrix computation.

δt	$\|K_1\|_2$	$\|K_2\|_2$	$\|K_4\|_2$	$\|K_6\|_2$
0.025	2.794×10^{-3}	4.280×10^{-5}	4.714×10^{-9}	8.777×10^{-14}
0.050	1.116×10^{-2}	3.420×10^{-4}	1.507×10^{-7}	1.122×10^{-11}
0.075	2.505×10^{-2}	1.152×10^{-3}	1.142×10^{-6}	1.915×10^{-10}
0.100	4.439×10^{-2}	2.724×10^{-3}	4.802×10^{-6}	1.432×10^{-9}
0.150	9.894×10^{-2}	9.120×10^{-3}	3.622×10^{-5}	2.433×10^{-8}
0.200	1.736×10^{-1}	2.138×10^{-2}	1.512×10^{-4}	1.809×10^{-7}

Table 12. Per-size error breakdown of every correction on the disordered TFIM. Global spectral-norm error at $t = 1$, $\delta t = 0.1$ ($J_i, h_i \sim \mathcal{U}[0.5, 1.5]$), mean $\pm$ one standard deviation over n_c disordered chains (sizes and n_c as in Table 8). $\epsilon_{w\leq 3}$ is the exact weight-≤ 3 oracle; $\epsilon^{\text{oracle}}_{\text{learned}}$ and $\epsilon^{\text{free}}_{\text{learned}}$ are the learned corrections trained on global-oracle and on oracle-free patch labels; ϵ_{BCH} is the analytic leading-order correction. Best learned column in each row is bold; sizes $n \geq 6$ are absent from training. All values recomputed from dense matrices.

n	regime	$\epsilon_{w\leq 3}$ (oracle)	$\epsilon^{\text{oracle}}_{\text{learned}}$	$\epsilon^{\text{free}}_{\text{learned}}$	ϵ_{BCH}
4	train	$6.526 \times 10^{-5} \pm 4.142 \times 10^{-5}$	$\mathbf{1.220 \times 10^{-4}} \pm 4.569 \times 10^{-5}$	$1.225 \times 10^{-4} \pm 4.475 \times 10^{-5}$	$1.166 \times 10^{-3} \pm 5.538 \times 10^{-4}$
5	train	$1.622 \times 10^{-4} \pm 9.451 \times 10^{-5}$	$2.253 \times 10^{-4} \pm 1.114 \times 10^{-4}$	$\mathbf{2.179 \times 10^{-4}} \pm 1.002 \times 10^{-4}$	$1.870 \times 10^{-3} \pm 8.674 \times 10^{-4}$
6	transfer	$1.999 \times 10^{-4} \pm 1.021 \times 10^{-4}$	$2.727 \times 10^{-4} \pm 9.416 \times 10^{-5}$	$\mathbf{2.616 \times 10^{-4}} \pm 9.702 \times 10^{-5}$	$2.125 \times 10^{-3} \pm 8.099 \times 10^{-4}$
7	transfer	$2.137 \times 10^{-4} \pm 1.267 \times 10^{-4}$	$3.280 \times 10^{-4} \pm 1.270 \times 10^{-4}$	$\mathbf{3.086 \times 10^{-4}} \pm 1.185 \times 10^{-4}$	$2.396 \times 10^{-3} \pm 9.617 \times 10^{-4}$
8	transfer	$2.532 \times 10^{-4} \pm 1.123 \times 10^{-4}$	$3.778 \times 10^{-4} \pm 9.838 \times 10^{-5}$	$\mathbf{3.621 \times 10^{-4}} \pm 1.126 \times 10^{-4}$	$2.823 \times 10^{-3} \pm 8.922 \times 10^{-4}$
9	transfer	$4.390 \times 10^{-4} \pm 1.578 \times 10^{-4}$	$\mathbf{5.533 \times 10^{-4}} \pm 1.601 \times 10^{-4}$	$5.570 \times 10^{-4} \pm 1.535 \times 10^{-4}$	$4.009 \times 10^{-3} \pm 1.121 \times 10^{-3}$
10	transfer	$4.951 \times 10^{-4} \pm 1.950 \times 10^{-4}$	$6.068 \times 10^{-4} \pm 1.706 \times 10^{-4}$	$\mathbf{5.814 \times 10^{-4}} \pm 1.945 \times 10^{-4}$	$4.542 \times 10^{-3} \pm 1.276 \times 10^{-3}$

Supplementary theory: finite-precision interpretation

The exact residual theorem is an algebraic identity. In floating-point arithmetic, the computed residual is $\widetilde{R}_q = \mathrm{fl}\{\widetilde{U}(\delta t)\widetilde{S}_q(\delta t)^\dagger\}$ and the computed corrected step is $\widetilde{G}_q = \mathrm{fl}\{\widetilde{R}_q\widetilde{S}_q\}$. Round-off enters through every matrix exponential and multiplication. This explains why the $q = 8$ baseline, already extremely accurate at $\delta t = 0.1$, can appear slightly more accurate than the oracle residual in Table 1: the theorem says the exact-arithmetic error is zero, while the table reports floating-point evaluations of different arithmetic circuits. For low and moderate order, the product-formula error dominates round-off and oracle RGTC gives many orders of magnitude improvement; for very high order at small step size, product-formula error can be below or comparable to double-precision round-off, so differences at 10^{-13} are not physically meaningful; projected residual experiments should be judged at error levels well above round-off, such as the $w = 3$ and $w = 4$ results in Extended Data Table 9.

Supplementary: reproducibility checklist

A reproducible implementation should verify $\left\|S_q^\dagger S_q - \mathbb{I}\right\|_2 < 10^{-12}$ for every tested q, n, and δt; verify $\left\|R_q^\dagger R_q - \mathbb{I}\right\|_2 < 10^{-12}$ for every oracle residual; verify $\|R_q S_q - U(\delta t)\|_2 < 10^{-12}$ for small and moderate orders; save raw CSV or JSON data before plotting; make plots only from saved raw data, not from hard-coded arrays; record package versions and BLAS backend information; use deterministic parameter grids or fixed random seeds; and include tests that fail if a citation key in `main.tex` is missing from `refs.bib`. These checks are especially important if the code is ported to GPU backends, where matrix-exponential implementations and floating-

Table 13. Single-step error versus step size for the learned residual at $n = 6$. Disordered open-boundary TFIM with $J_i, h_i \sim \mathcal{U}[0.5, 1.5]$; each row is a mean ± one standard deviation over $n_c = 24$ disordered chains. $\epsilon_{\mathrm{Strang}}$ is the uncorrected per-step error; $\epsilon_{w\le 3}$ the exact weight-≤ 3 oracle; $\epsilon_{\mathrm{learned}}$ the oracle-free learned correction; ϵ_{BCH} the analytic leading-order (δt^3) correction. As δt grows the analytic baseline degrades toward the uncorrected step while the δt-conditioned learned correction tracks the oracle, quantifying Fig. 5b. Best (smallest) correction in each row is bold; values recomputed from dense matrices.

δt	$\epsilon_{\mathrm{Strang}}$	$\epsilon_{w\le 3}$ (oracle)	$\epsilon_{\mathrm{learned}}$	ϵ_{BCH}
0.05	$6.269 \times 10^{-4} \pm 1.349 \times 10^{-4}$	$\mathbf{7.379 \times 10^{-7}} \pm 3.183 \times 10^{-7}$	$9.201 \times 10^{-6} \pm 3.947 \times 10^{-6}$	$5.462 \times 10^{-5} \pm 1.511 \times 10^{-5}$
0.10	$5.195 \times 10^{-3} \pm 1.225 \times 10^{-3}$	$\mathbf{2.397 \times 10^{-5}} \pm 1.195 \times 10^{-5}$	$4.567 \times 10^{-5} \pm 1.795 \times 10^{-5}$	$1.200 \times 10^{-3} \pm 3.934 \times 10^{-4}$
0.15	$1.684 \times 10^{-2} \pm 6.207 \times 10^{-3}$	$\mathbf{1.903 \times 10^{-4}} \pm 1.582 \times 10^{-4}$	$2.456 \times 10^{-4} \pm 1.685 \times 10^{-4}$	$6.338 \times 10^{-3} \pm 3.271 \times 10^{-3}$
0.20	$4.407 \times 10^{-2} \pm 1.139 \times 10^{-2}$	$\mathbf{8.838 \times 10^{-4}} \pm 4.680 \times 10^{-4}$	$1.042 \times 10^{-3} \pm 5.405 \times 10^{-4}$	$2.382 \times 10^{-2} \pm 8.645 \times 10^{-3}$
0.25	$7.622 \times 10^{-2} \pm 2.300 \times 10^{-2}$	$\mathbf{2.135 \times 10^{-3}} \pm 1.122 \times 10^{-3}$	$2.596 \times 10^{-3} \pm 1.502 \times 10^{-3}$	$5.098 \times 10^{-2} \pm 2.057 \times 10^{-2}$
0.30	$1.347 \times 10^{-1} \pm 3.801 \times 10^{-2}$	$\mathbf{6.044 \times 10^{-3}} \pm 3.522 \times 10^{-3}$	$7.684 \times 10^{-3} \pm 4.756 \times 10^{-3}$	$1.111 \times 10^{-1} \pm 4.106 \times 10^{-2}$

point accumulation order may differ from CPU reference values.

Supplementary: possible extensions

The residual-generator framework suggests several concrete research extensions. A *BCH residual compiler* derives explicit nested-commutator formulas for K_q through a target order and implements $e^{-iK_q^{\mathrm{BCH}}}$ as a short product formula; Theorems 4 and 6 convert truncation error into global simulation error. A *variational residual compiler* uses the oracle residual on small systems as a target for a parameterized local circuit $V(\theta)$, with loss $\|V(\theta) - R_q\|_F^2$, local Pauli-shadow loss, or state-averaged infidelity, and the stability theorem provides the operator-norm target. An *operator-learning residual compiler* is realized in Sec. 1.9 for the Strang TFIM, learning the step-size-conditioned local weight-≤ 3 coefficients of K_2 with a single size-transferable network trained from oracle-free local patches; natural continuations extend the local templates to other lattices and higher orders, replace patch labels with classical-shadow or commutator estimates on hardware, and compile the learned generator $e^{-i\widehat{K}_q}$ into native gates using Theorems 4 and 6 as the operator-norm target. A *GPU-accelerated benchmark generator* uses CPU dense simulation as a correctness reference, then ports parameter sweeps and Pauli projections to batched GPU kernels, keeping output tolerance-comparable to the CPU reference and always regenerating raw data before producing figures.